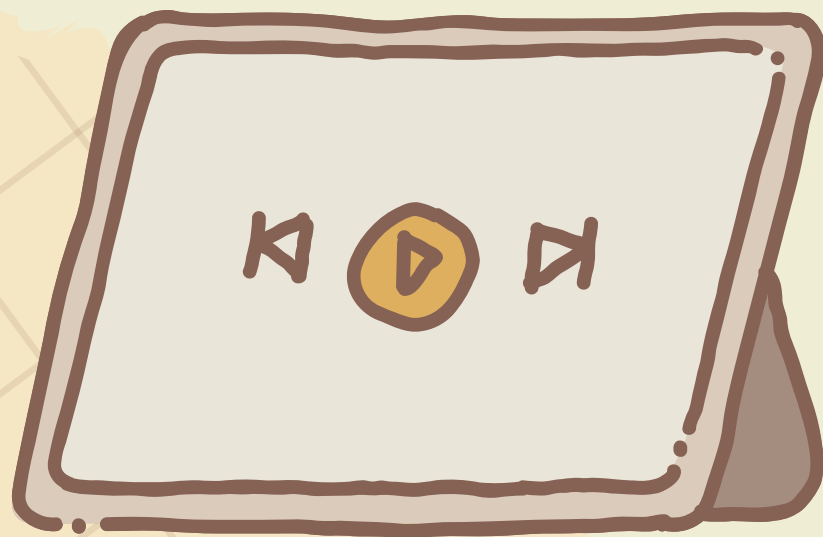




Soundtrack to Success

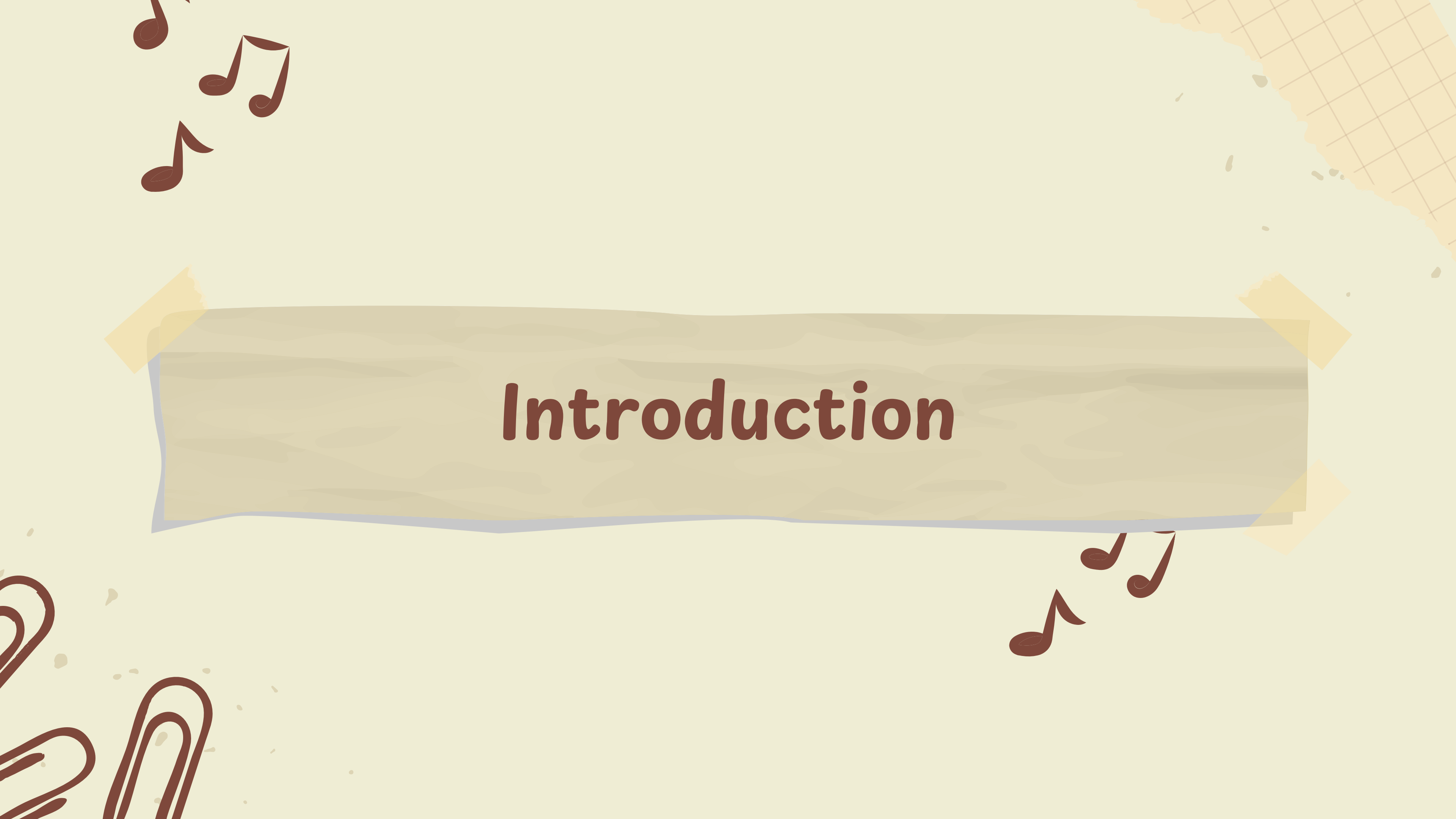
By: Musivation





List of Contents

1. Introduction
2. Music & Motivation
3. Neurobiology
4. Methods
5. Results & Analysis
6. Discussion
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Introduction

What is motivation?

**The thing that gives purpose or direction to behavior
can be conscious or unconscious**

(American Psychological Association)

Intrinsic

Led by curiosity, personal
interest, and personal
fulfillment

Extrinsic

Led by outside rewards,
recognition, or avoiding
negative consequences

What is learning?

Learning is

- A change in mental states that is associated with some environmental or cognitive event
- People learn to develop a better understanding of things rather than just memorization
- Students begin as novice learners and build their way up to expert level on various topics and concepts

(Bransford et al. 2004)

Word list task

- All forms of memory are used in language processing short-term --> long-term
- Word lists work best for younger people
- "Learning is considered equal to the storing, retaining, and recalling information"
- Semantically relevant words are more likely to be remembered and recalled

(Khosravizadeh et al 2011)

Motivation to Learn

Studies In Children Learning Choice

- Self Motivated (interest)
 - Authority Figure
 - Teachers
 - Family
- Reading as a shared experience
 - Children happy to receive books as gifts
 - Children regularly read to young likely to take independent interest

Choice

- Self-selection and agency
- Extrinsic motivators less effective

College Students

- Subject authority figure gap
- Authority figure motivates completion, not quality
- Bare minimum

(Edmunds, Bauserman 2006)



The image features a central, horizontal, rectangular piece of light brown paper with a slightly textured, watercolor-like appearance. This paper is pinned to a light cream-colored background with four yellow, torn-edge adhesive tabs at its corners. The title "Music & Motivation" is written in a bold, dark brown, sans-serif font across the center of the paper. Surrounding the paper are several hand-drawn musical notes in a dark brown color. In the top-left corner, three notes are clustered together. In the bottom-right corner, two notes are visible. In the bottom-left corner, there are two large, stylized outlines of what appear to be headphones or a similar audio-related device. The overall aesthetic is casual and creative, resembling a student's project or a hand-drawn poster.

Music & Motivation

Why Music?

- The auditory sense is one that cannot be easily controlled or focused on a certain thing
 - Can lead to many distractions
- People use headphones and music for “personal soundscape curation”
 - Giving them the opportunity to control their auditory environment and the signals being taken in
 - Can improve focus and direct attention in a work environment

(Haas et al., 2018)





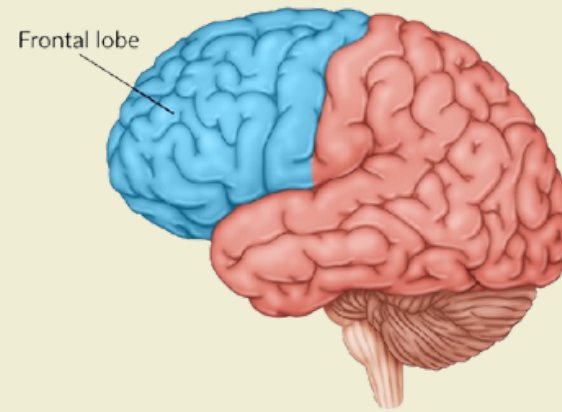
Neurobiology

Brain Regions Activated

Music affects functional connectivity in the brain

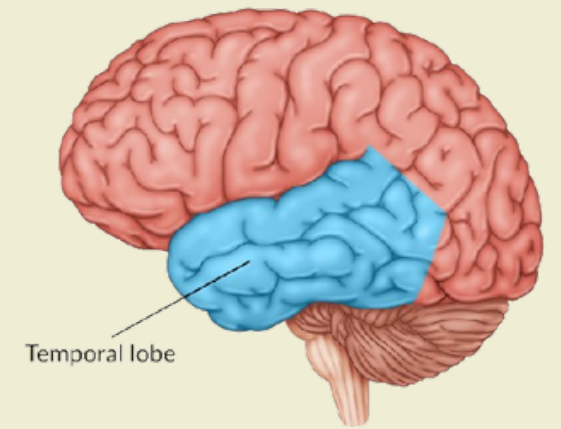
Frontal Lobe

- Increased connectivity when listening to relaxing music
- Long-term exposure (30 mins a day for 2 weeks) resulted in stronger frontal lobe connections



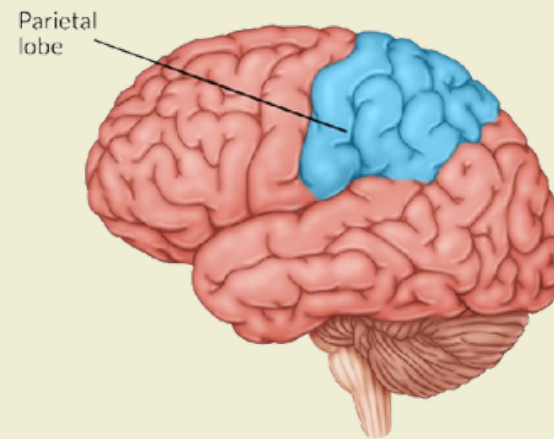
Temporal Lobe

- Increased connectivity when listening to favorite music
- Stronger emotion and memory component



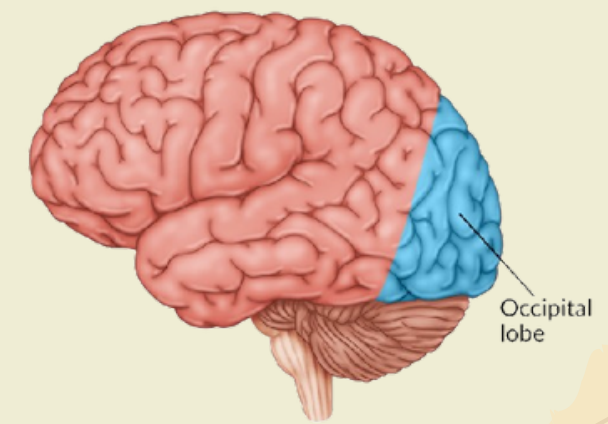
Parietal Lobe

- Increased connectivity when listening to relaxing music
 - Findings suggested better cognitive engagement and memory retention when music was incorporated into study breaks



Occipital Lobe

- Beta band activity was affected
- Suggests that music might influence the processing of visual information



(Mahmood et al., 2022)

System Overload

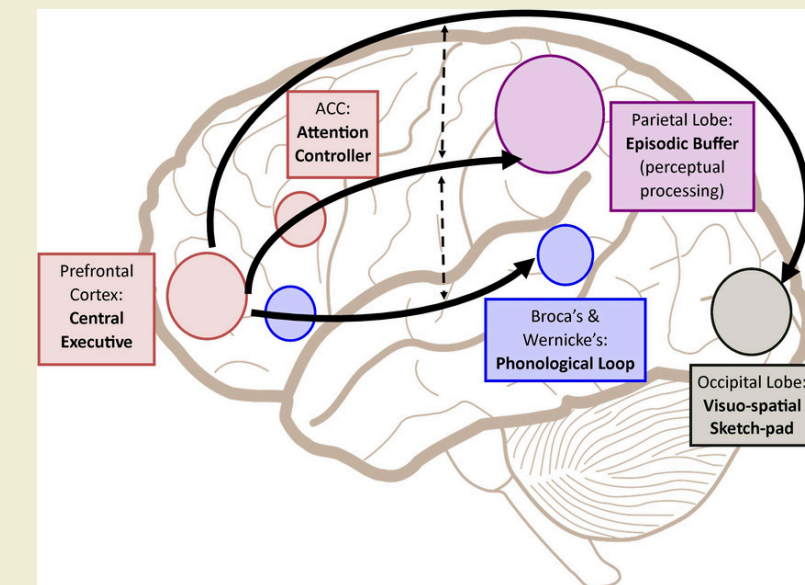
Lyrical Impact

- Music with lyrics had a more detrimental effect on performance than instrumental music, dual task interference
- Presence of lyrics, adds to the cognitive load which makes it harder to focus
- Lyrics may be less disruptive in math or problem solving tasks, and more disruptive in language based tasks

(Dolegui, 2013)

Phonological Loop

- Music with lyrics may have a semantic or phonological interference on given tasks
- People perceive music with lyrics as generally distracting



(Souza and Barbosa, 2023)

Your Brain On Music

What is activated?

The Role

Music and intrinsic motivation!

Amygdala

- Higher connection to nucleus accumbens during joyful music
- Superficial amygdala (reward relating to emotional signals)

- Can contribute to approach-withdrawal behaviors based on music cues given
 - Linking environment cues to motivational outcomes

Nucleus Accumbens

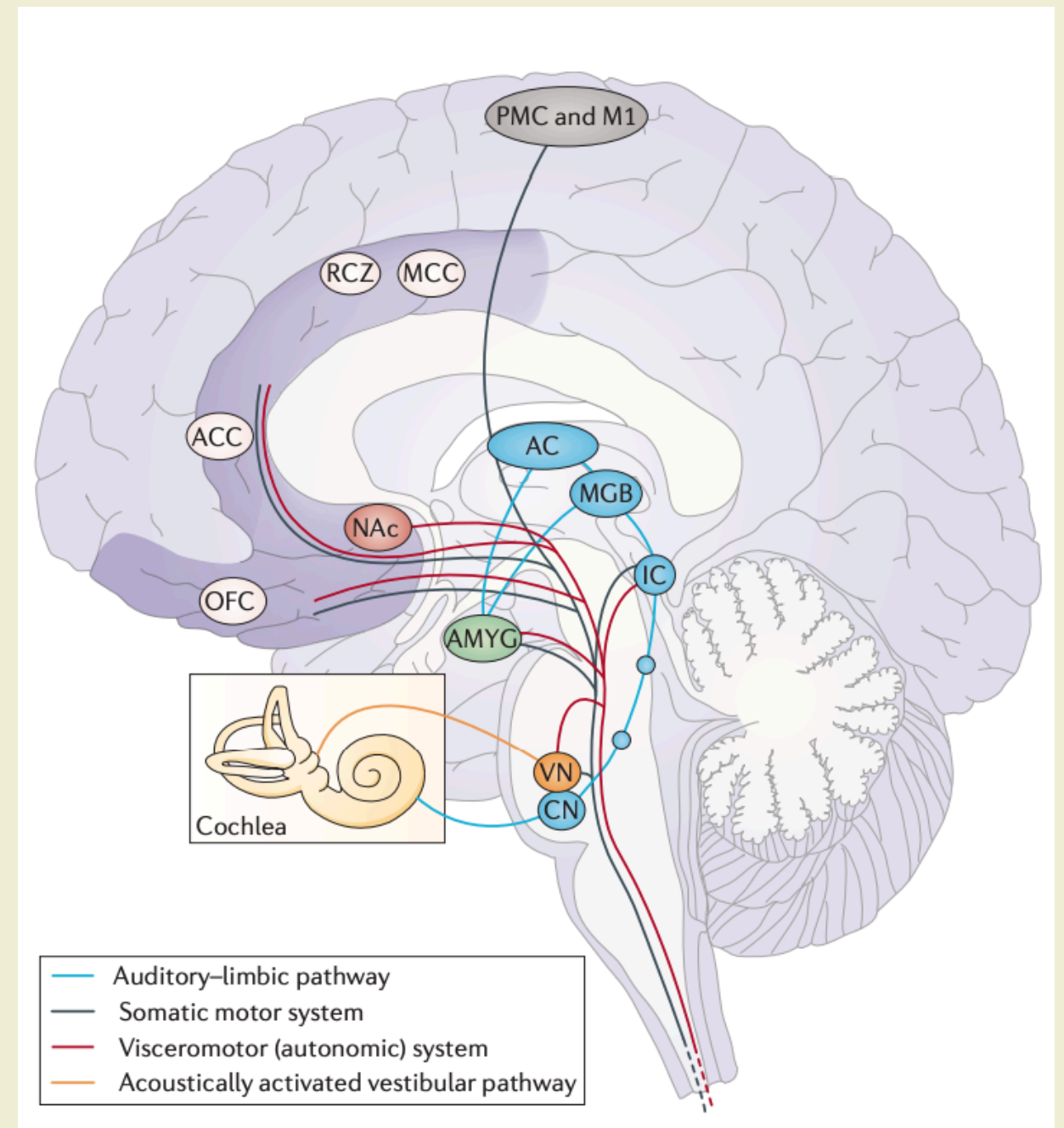
- Evoked by music connected to pleasure and reward
- Motivates reward attainment

- Contributes to the activation of reward networks (function to ensure survival of a species)

VTA

- Positive music activates the mesolimbic dopaminergic reward pathway

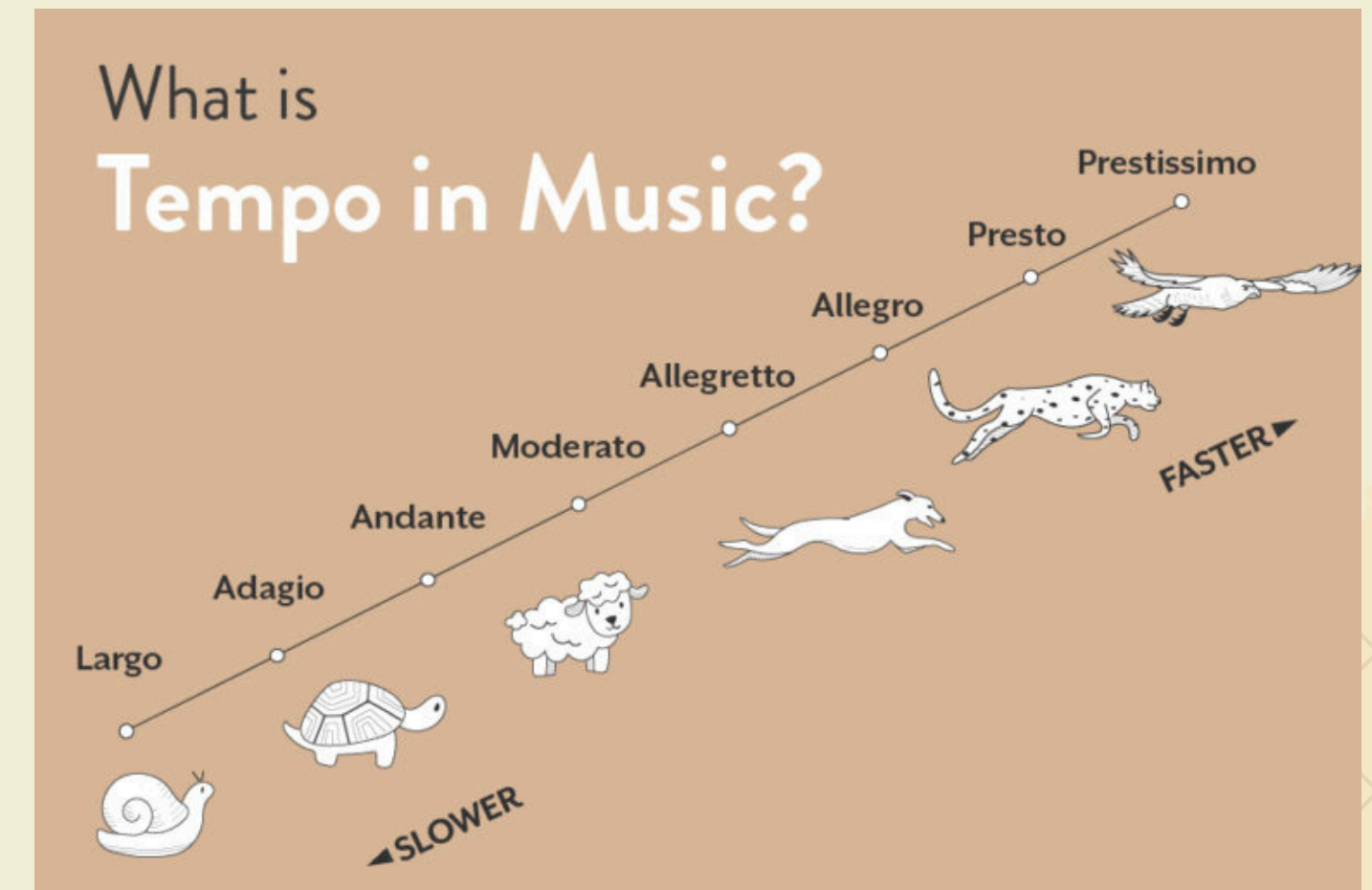
- Anticipates a reward and prepares for the rewarding experience
- Contributes to motivation when seeking out rewarding behaviors



(Koelsch, 2014) (Vuilleumier & Trost, 2015)

What about Tempo?

- Faster tempo music can improve accuracy in difficult tasks involving decision-making, but not typically in tasks with “complex cognitive effort”
- Slow tempo music can negatively influence cognitive performance as it decreases processing speed
 - Can be helpful in tasks with no time limits that are more complex
- There can be benefits to playing music prior to doing a task as it can impact a person’s cognitive processing speeds
 - Faster music before more simple tasks
 - Slower music before more complex tasks (supports accuracy)



(Lin et al., 2023)



Research Questions

To what extent does background music influence task performance and motivation to learn?

Does the type of music play a role in these effects?

Hypothesis

- Non-lyrical music will better support learning compared to lyrical music
- Lyrical music will have slower reaction times as it diminishes focus
(competition with cognitive resources)
- Faster tempo songs will be more beneficial for a quick learning task
- Silence will provide a baseline for measuring the effects





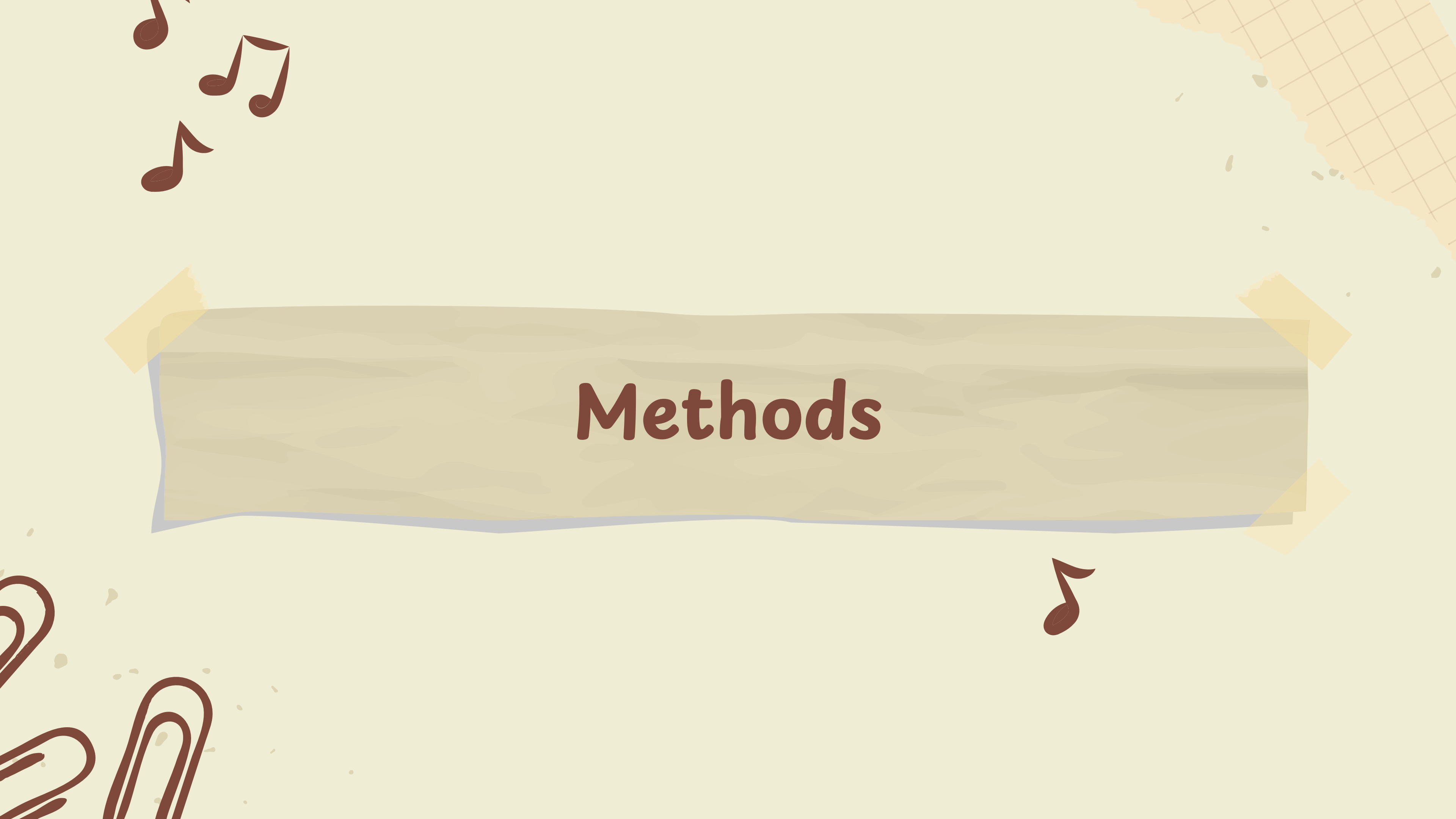
Goals & Objectives

Effective Learning

- Investigate how different types of music influence motivation while studying and getting work done
 - Examine how music presence vs. silence affects task accuracy and trial performance
 - Determine whether engagement varies across different learning conditions
- 

Genre & Tempo Influence

- Explore whether tempo affects motivation and cognitive performance
 - Compare different genres to identify styles that enhance or hinder learning
 - Investigate how faster or slower tempo songs affect response time and accuracy during tasks
- 
- 



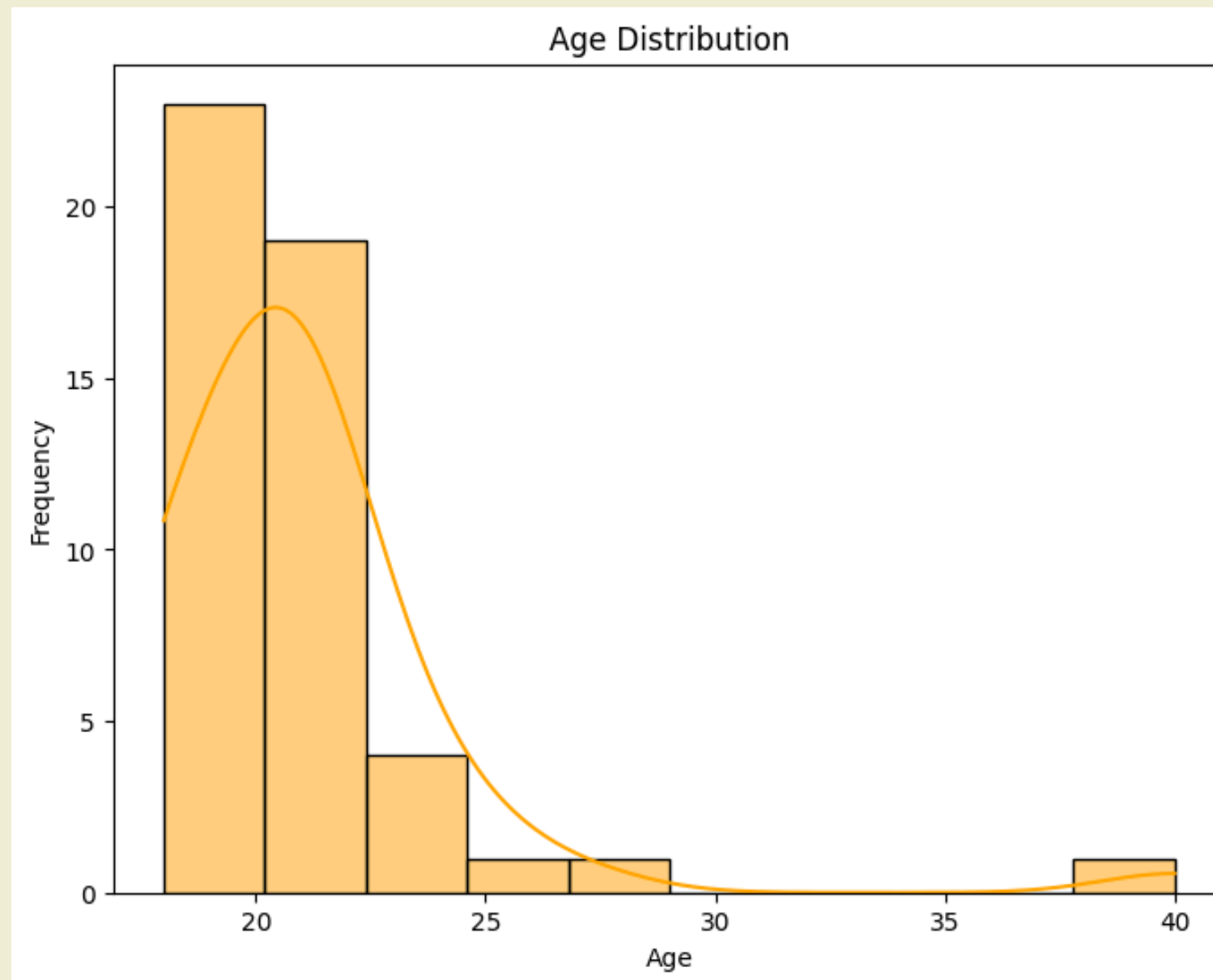
Methods

Study Design

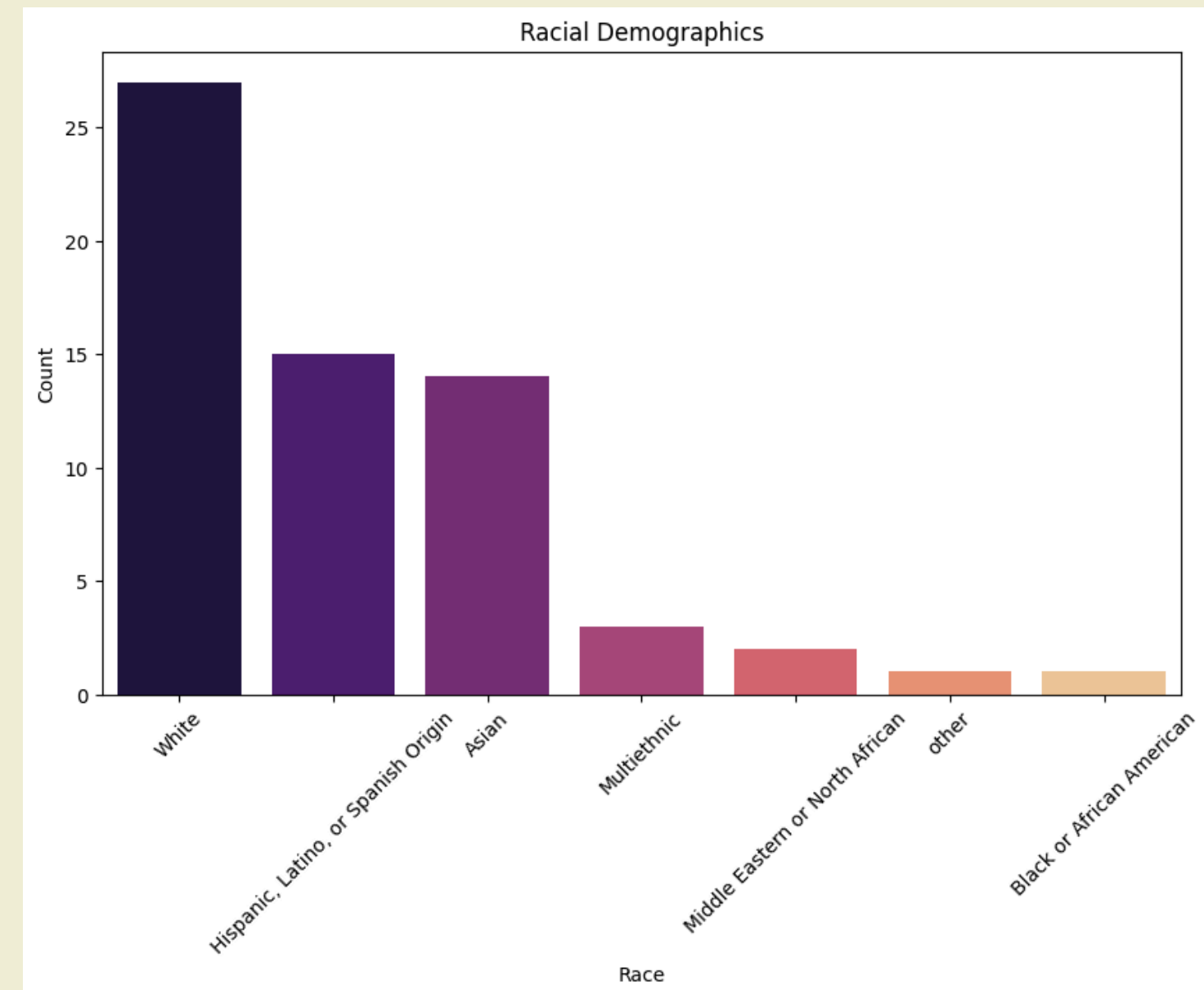
- Within-subjects design with Latin Square counterbalancing to balance condition exposure and control for order effects
- Built using the [jsPsych v8](#) JavaScript framework for browser-based behavioral experiments
- Hosted with [GitHub Pages](#) for ease of access and cross-platform compatibility
- Recruited **49** participants virtually through texts, Discord, etc., and through QR codes during lectures
- Collected data using Open Science Framework's [DataPipe](#) and exported as CSV files



Participants



- The average age of participants was 21
- Participant ages ranged from 18 to 40



- Participants were predominantly White, of Hispanic / Latino / Spanish origin, and Asian

Experiment Overview

Baseline Evaluation

Eight question survey used to collect demographics and assess study habits, baseline motivation, and music preference



Learning Trials

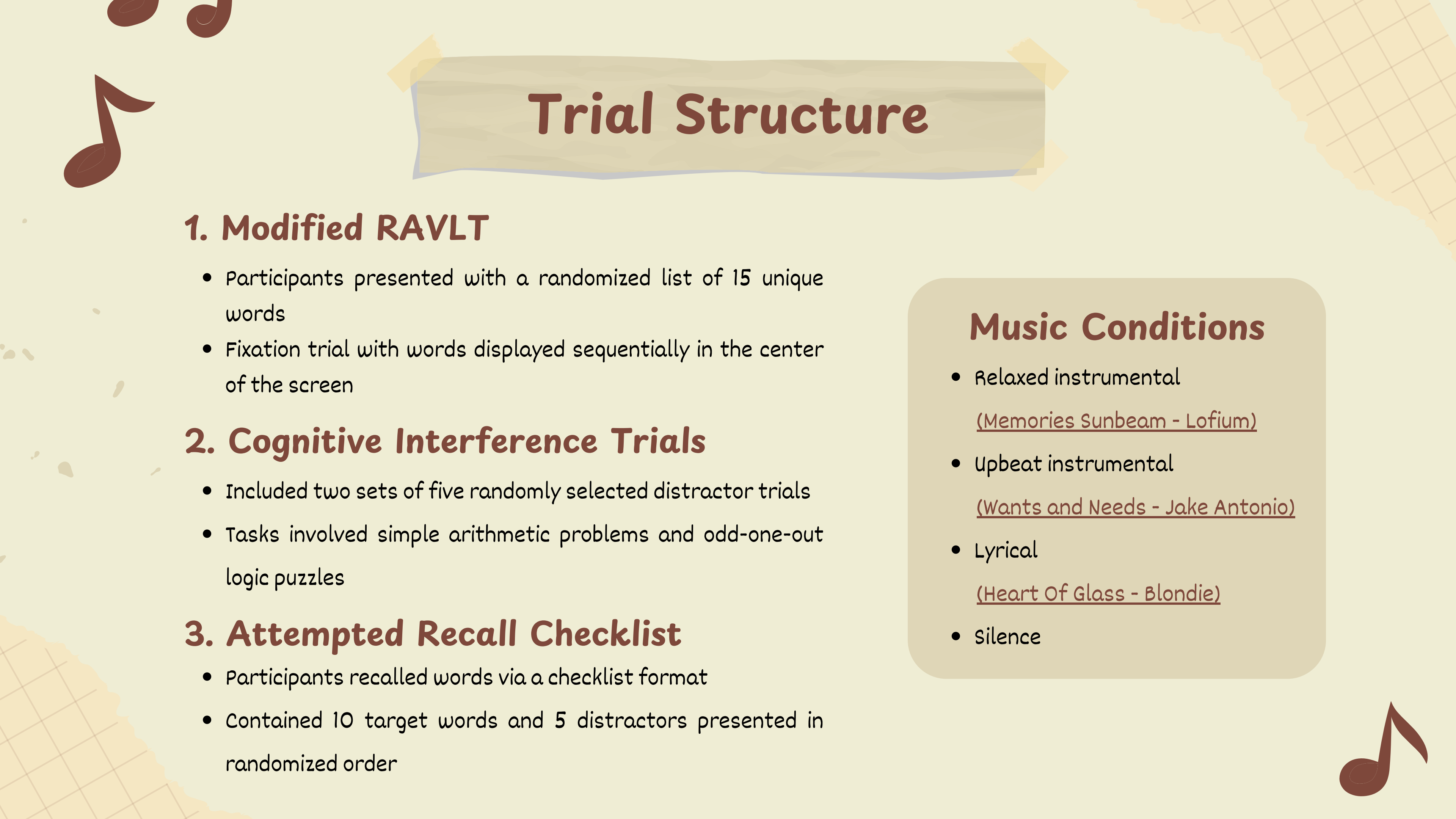
Four series of trials including verbal recall, math problems, and odd-one-out logic puzzles under different music conditions



Reflection Survey

Ten question survey reflecting on mood, motivation, and perceived impact of each music condition on task performance





Trial Structure

1. Modified RAVLT

- Participants presented with a randomized list of 15 unique words
- Fixation trial with words displayed sequentially in the center of the screen

2. Cognitive Interference Trials

- Included two sets of five randomly selected distractor trials
- Tasks involved simple arithmetic problems and odd-one-out logic puzzles

3. Attempted Recall Checklist

- Participants recalled words via a checklist format
- Contained 10 target words and 5 distractors presented in randomized order

Music Conditions

- Relaxed instrumental
(Memories Sunbeam - Lofium)
- Upbeat instrumental
(Wants and Needs - Jake Antonio)
- Lyrical
(Heart Of Glass - Blondie)
- Silence

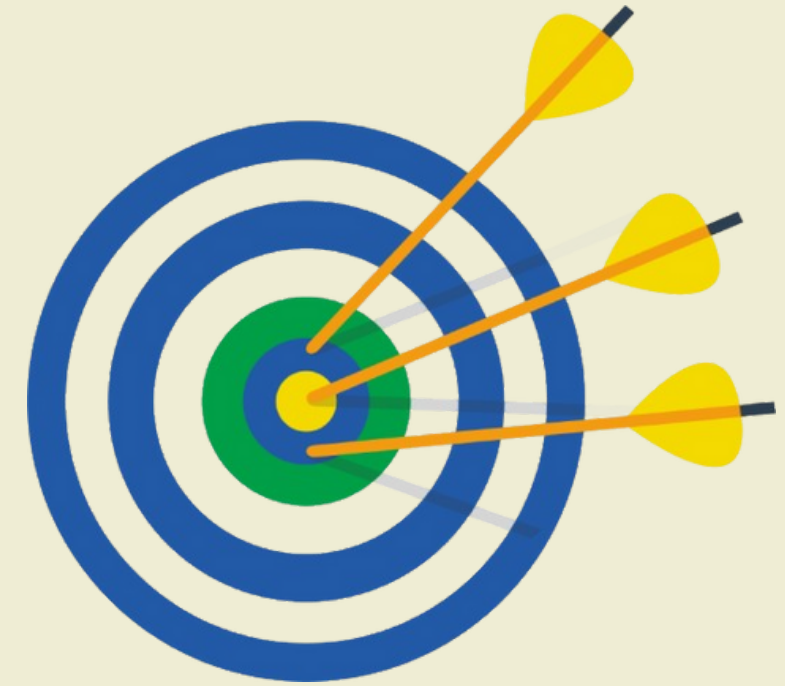
Key Measures

Learning Trial Performance

- Task Accuracy: Proportion of correctly selected responses during trial
- False Alarms: Incorrectly selected words from the checklist
- Miss Rate: Target words that were not selected during recall

Motivation Metrics

- Response Time: Time taken to complete each trial
- Motivation Levels: Self-reported baseline and post-task motivation
- Condition Ranking: The perceived impact of music conditions on motivation and performance

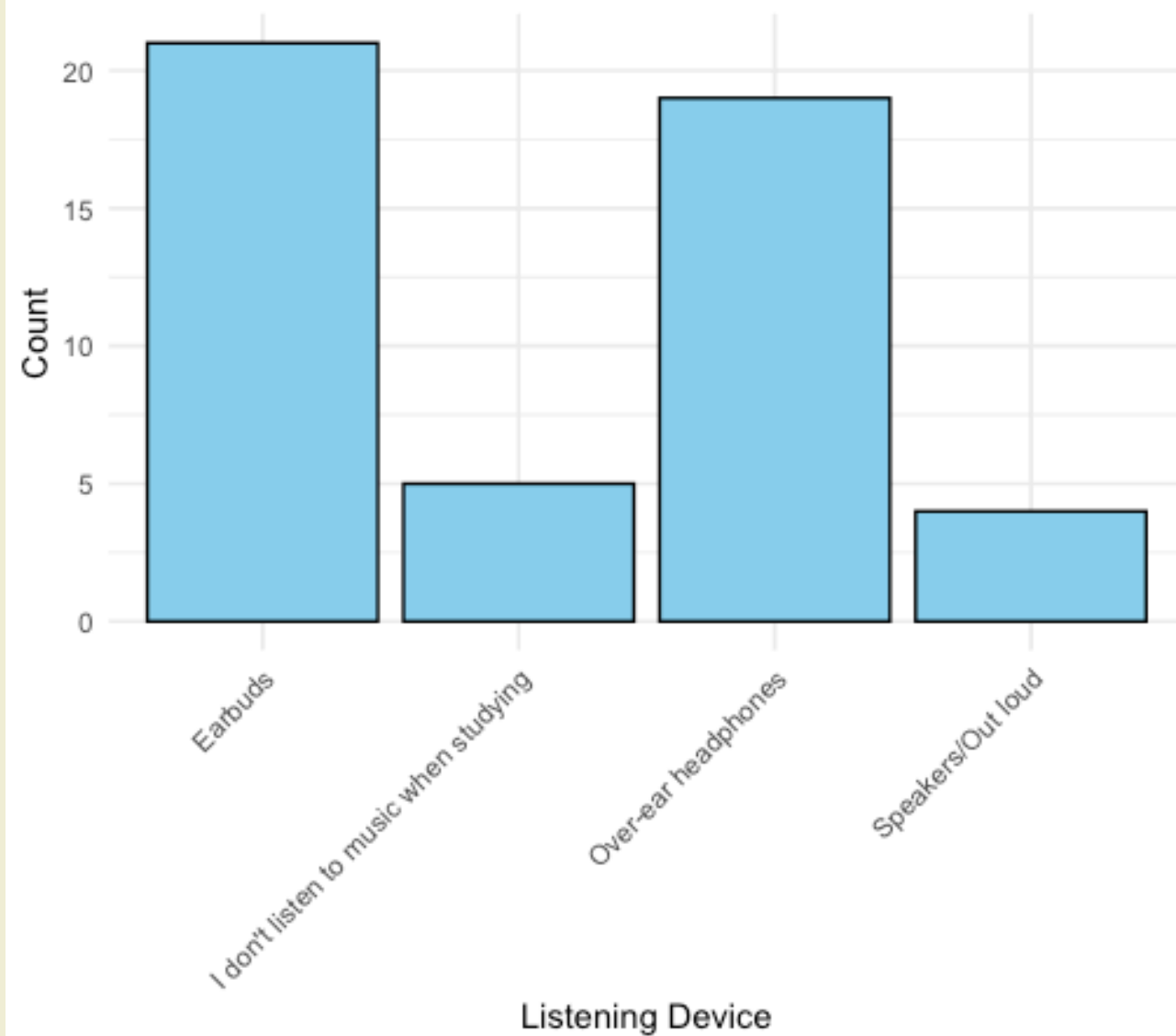




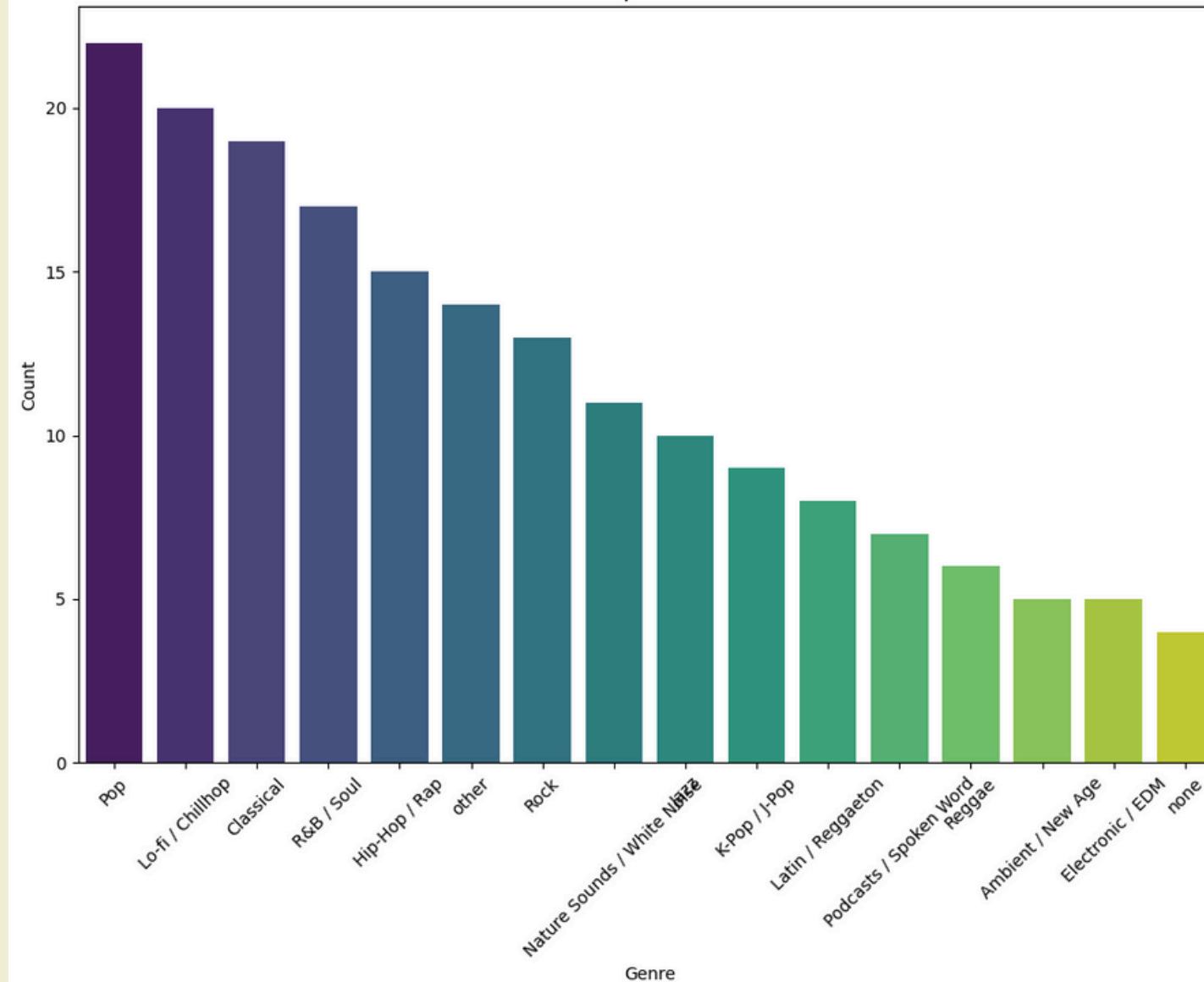
Results & Analysis

Results

Distribution of Baseline Listening Device



Most Popular Genres



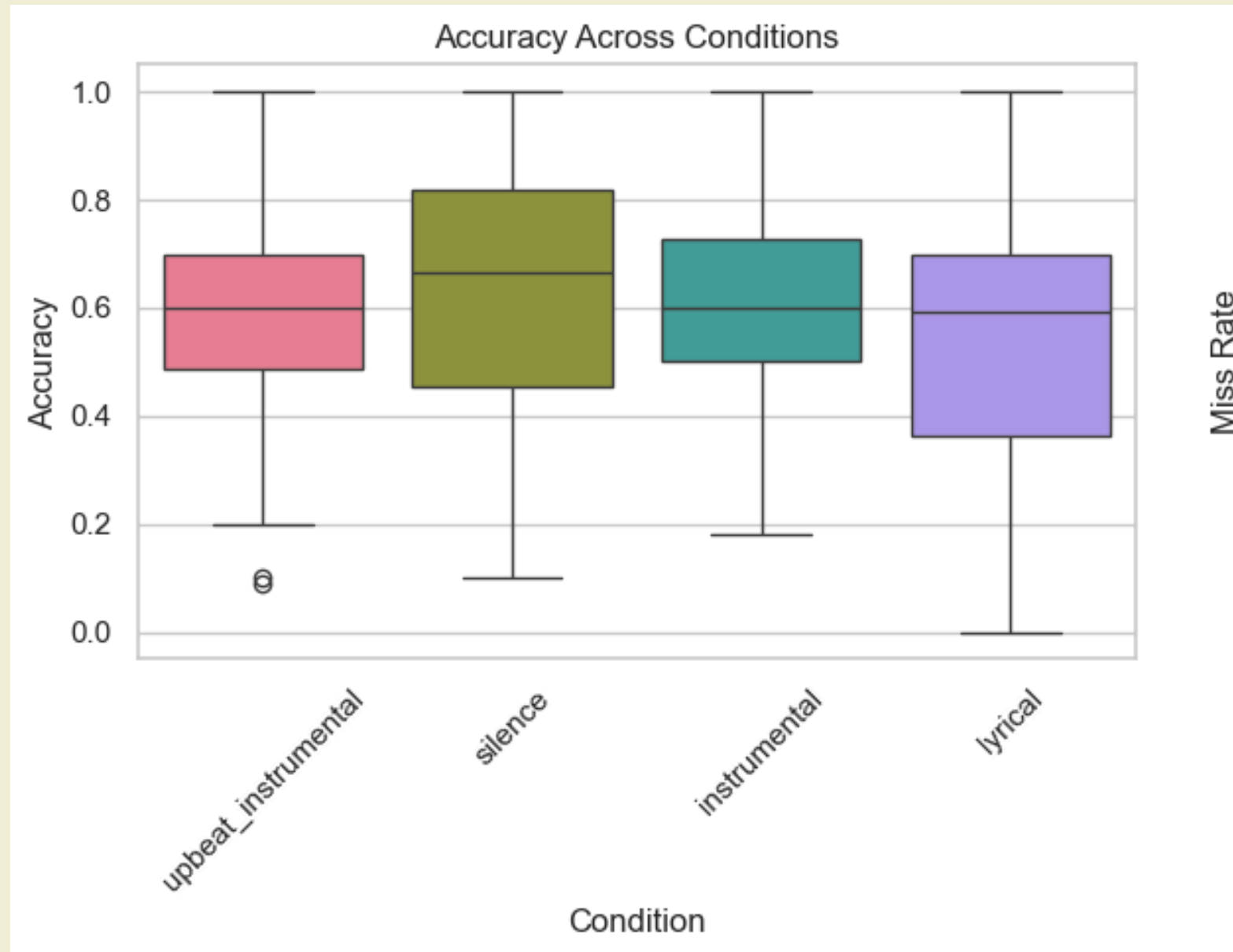
Top Genres

1. Pop
2. Lo-fi
3. Classical
4. R&B / Soul
5. Hip-Hop / Rap

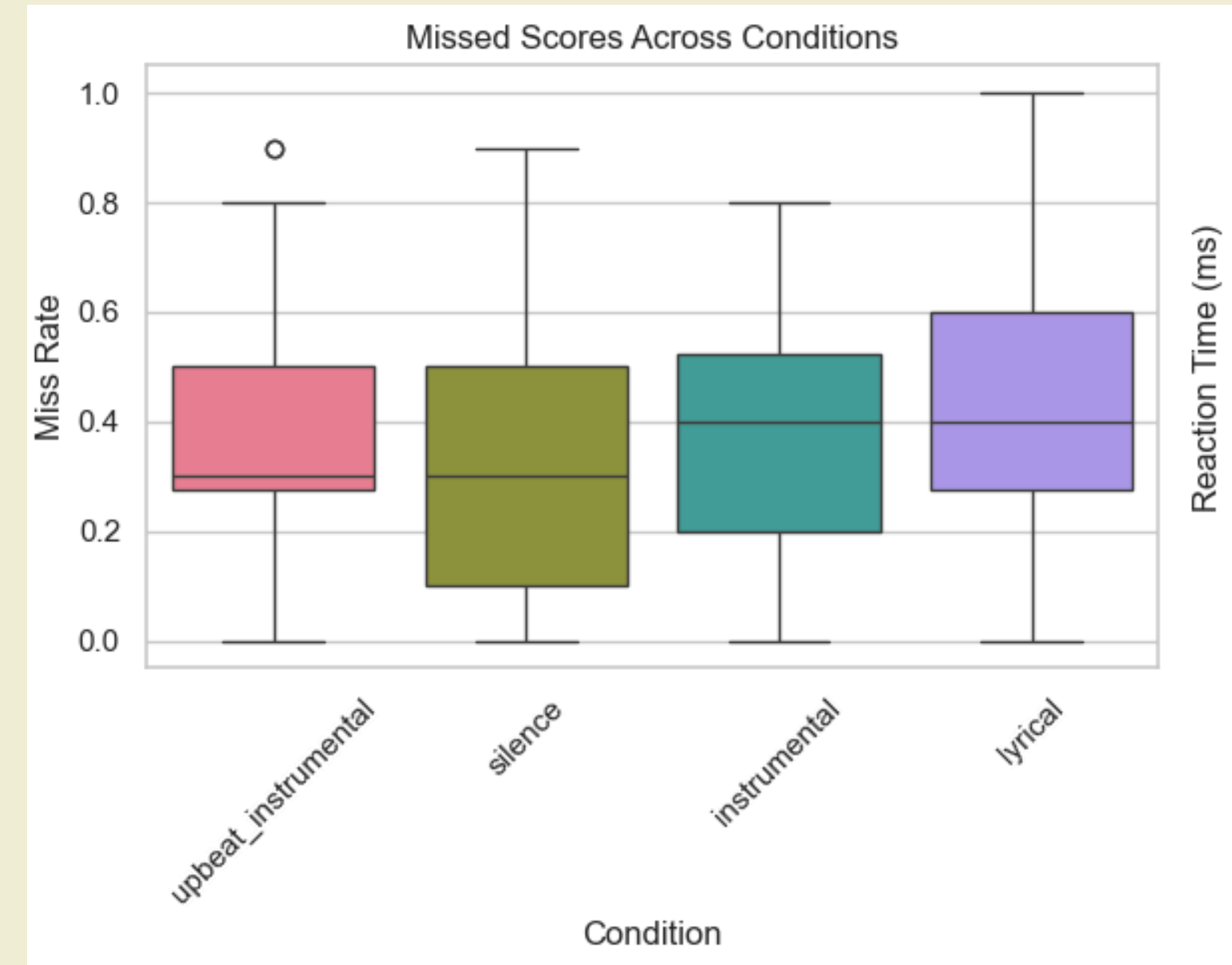
- Higher use of earbuds and over-ear headphones when listening to music

Results

*Lines are 25th, 50th, and 75th percentiles

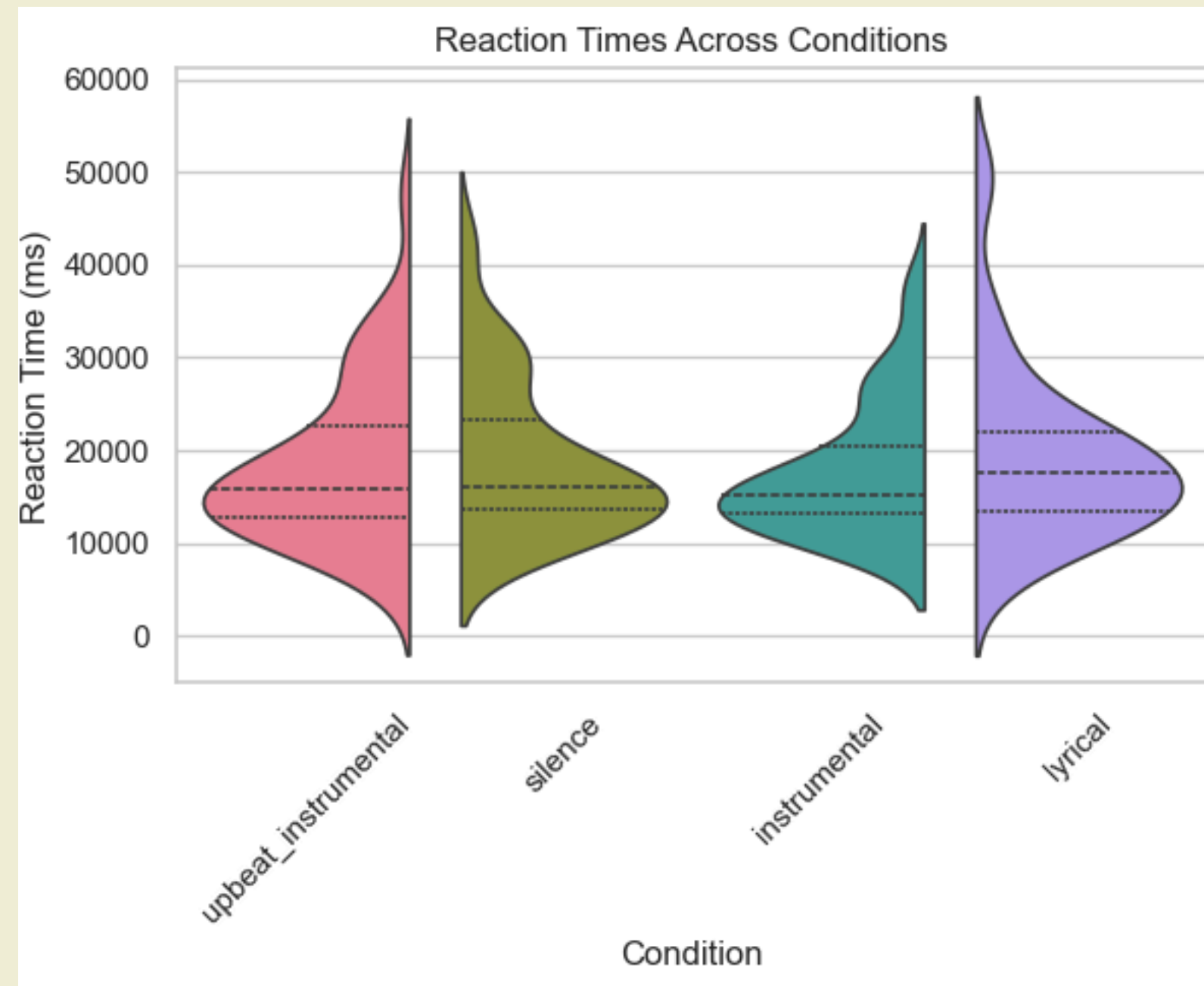


- Lowest general accuracy found during lyrical trial
- Silence had the best performance during the word-list learning task



- People tended to miss less while doing the silent trial
- High variation still seen among silence and lyrical trials

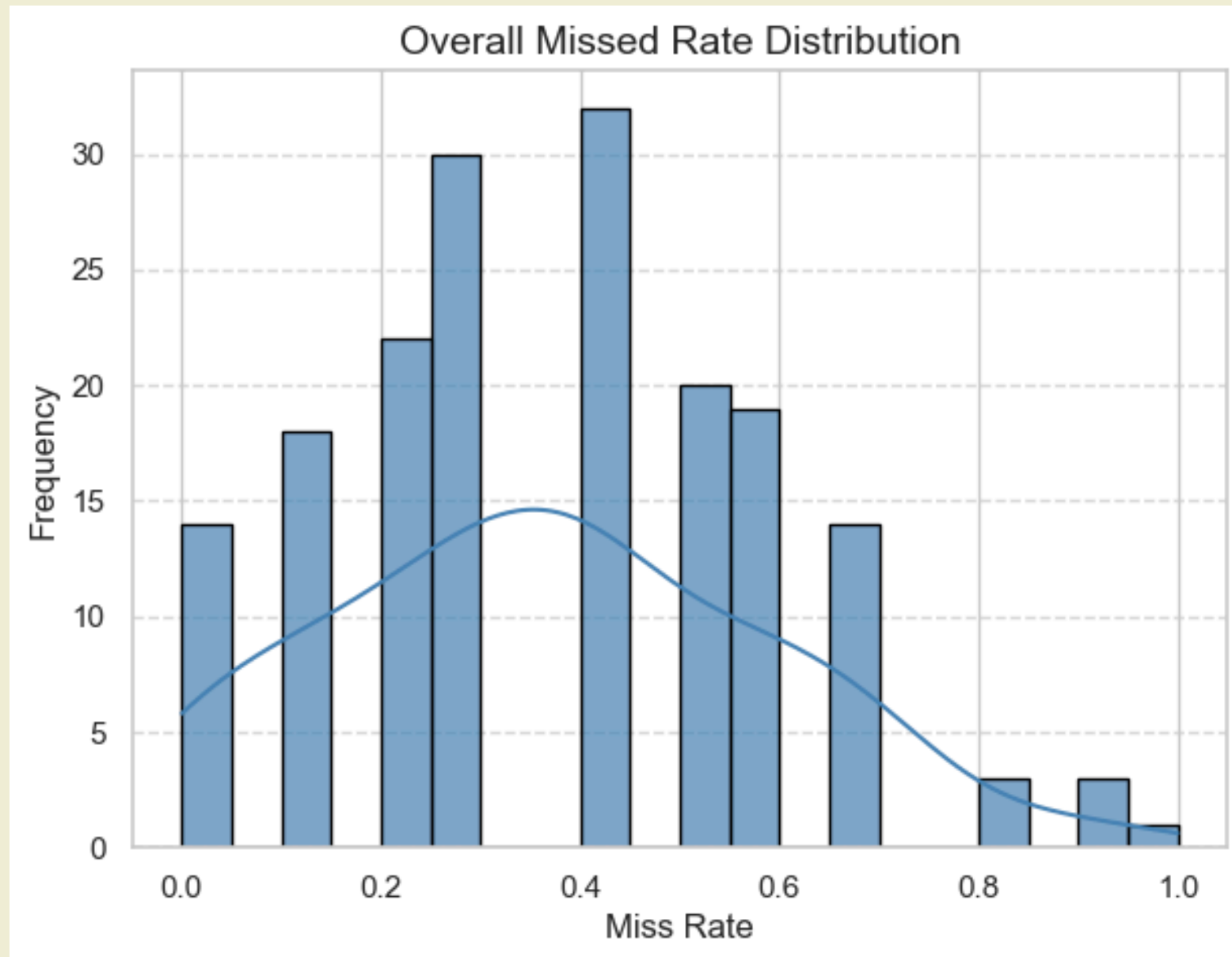
Results



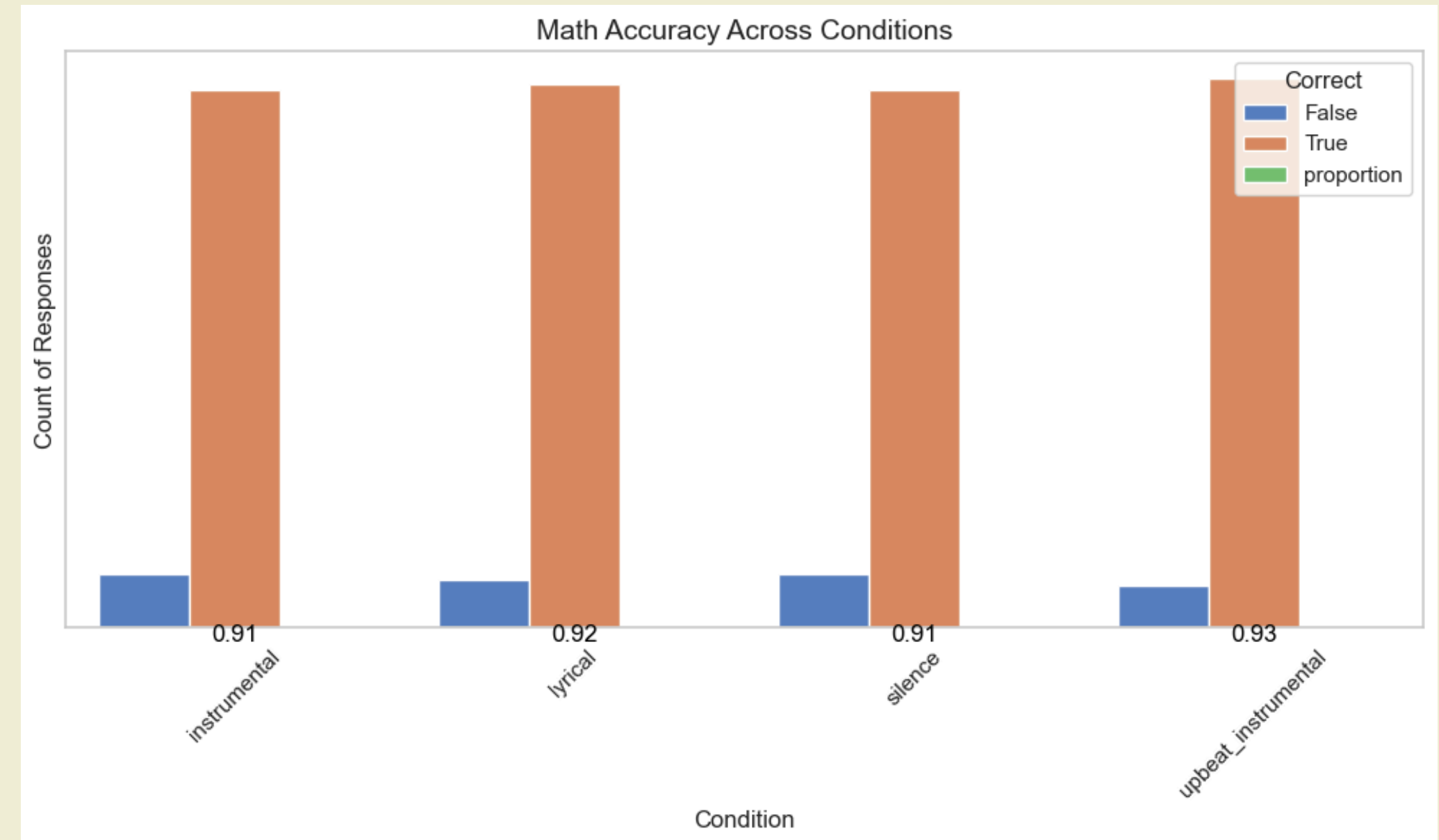
- Most variance found among lyrical trial
- Overall similar average reaction times
- Could be due to nature of the study

*Lines are 25th, 50th, and 75th percentiles

Results



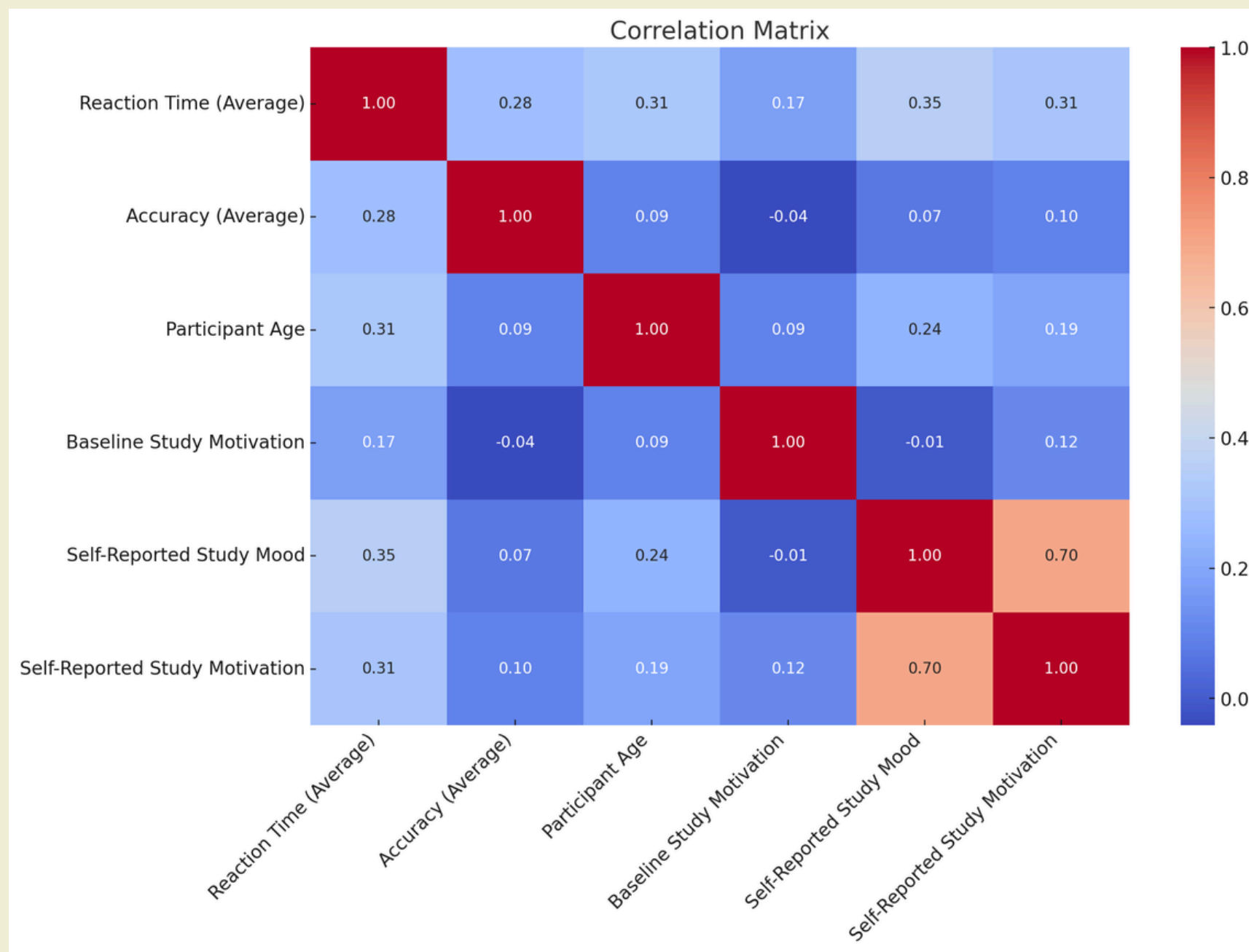
Average miss rate: 31%



Typically high rates of accuracy with math throughout all trials

- Could be impacted by brain regions being utilized to do math tasks compared to other learning tasks

Results



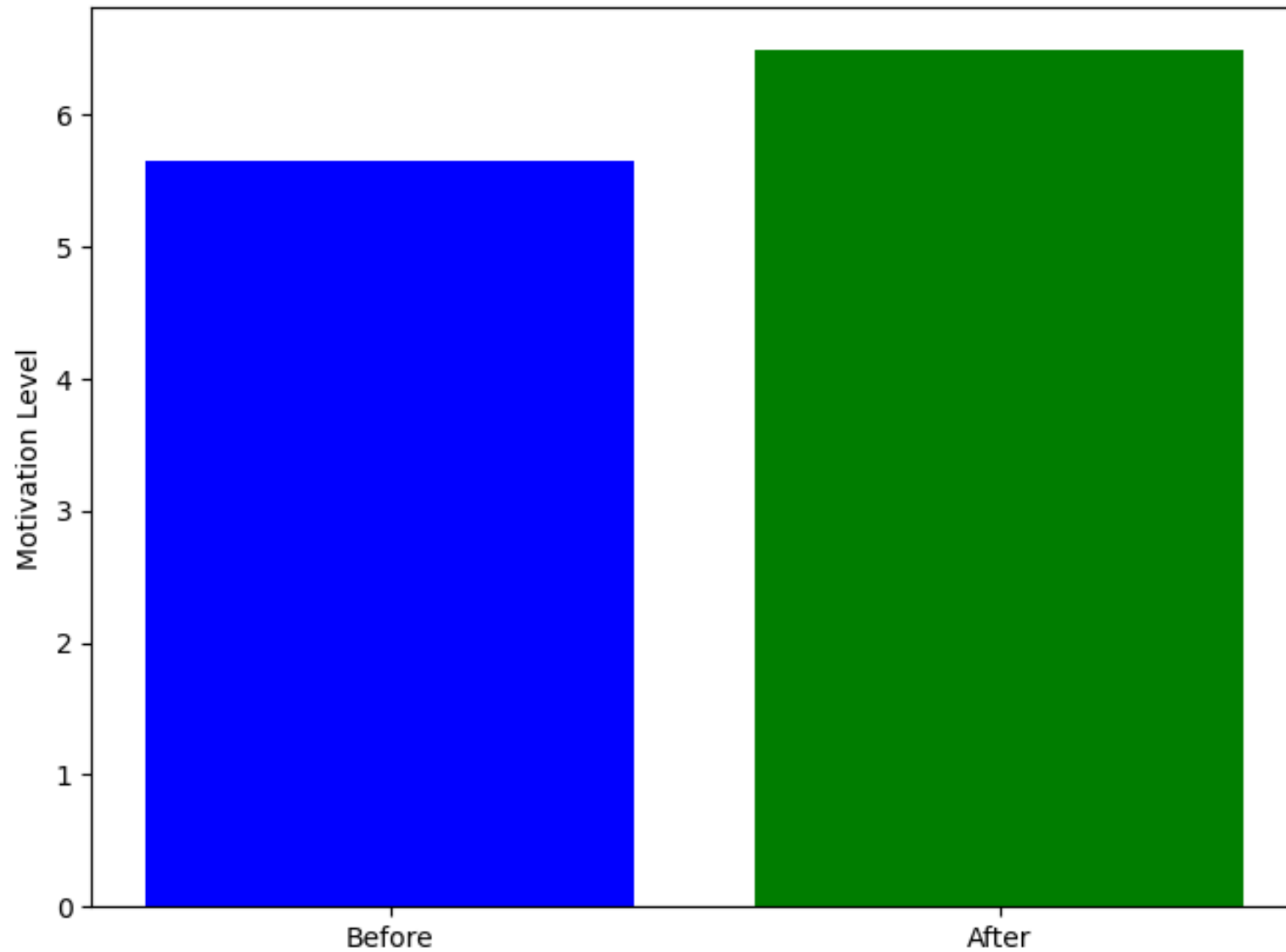
There was a moderate positive correlation (0.70) between self-reported mood and motivation following completion of the study.

This Suggests

- Participants who reported a better mood also tended to report higher motivation
- Students who feel good while studying are more motivated, or vice versa

Results

Average Motivation Levels Before and After Study



Average Pre-Motivation: 5.65

Average Post-Motivation: 6.49

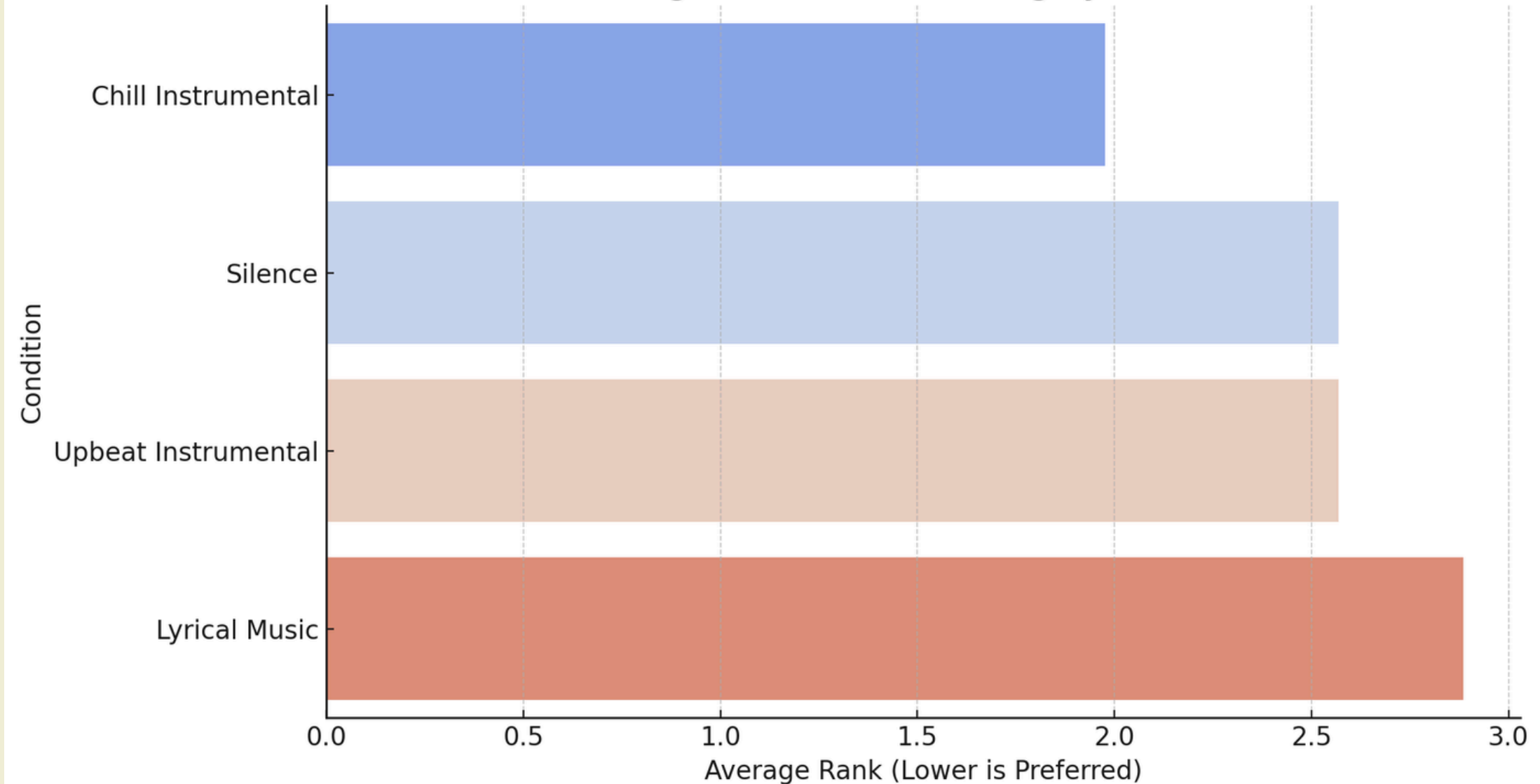
Paired t-test:

($t(49) = -2.22, p < 0.05$)

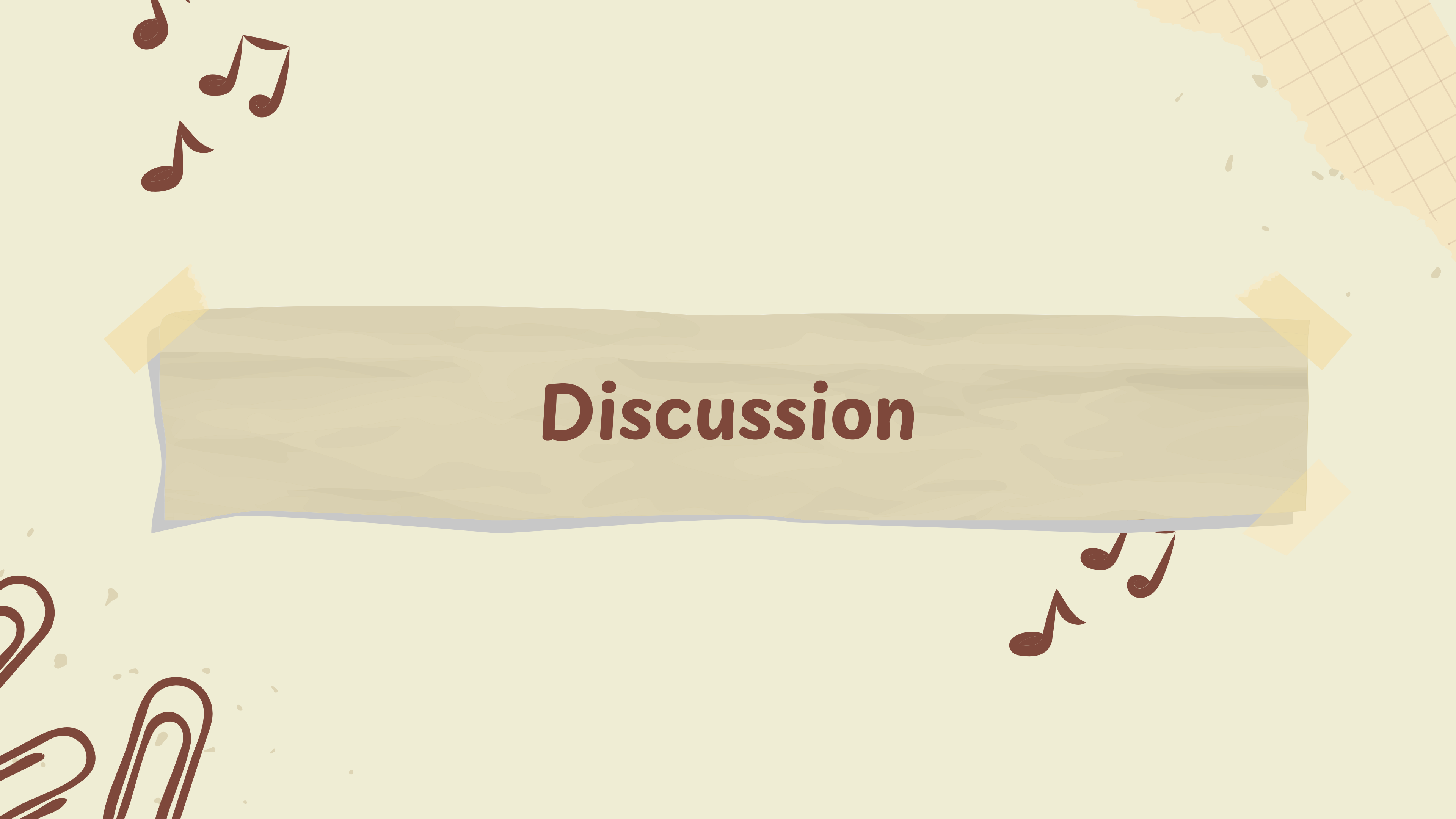
Mood and Post-Motivation:

($t(49) = -4.88, p < 0.05, p < 0.01$)

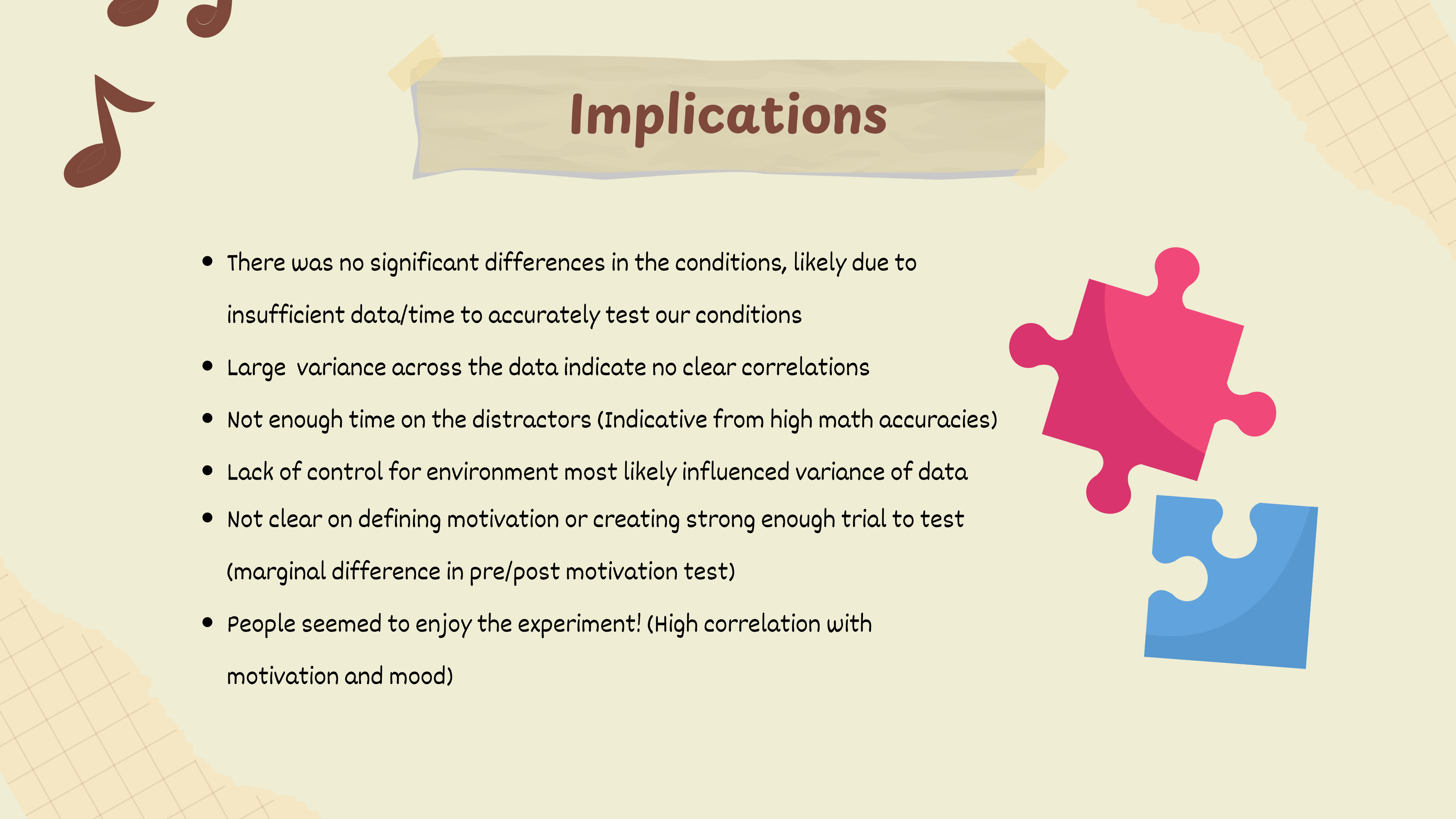
Average Motivation Ranking by Condition



- Chill Instrumental had the lowest average ranking (1.98), making it the most preferred condition for motivation
- Even if performance was better with silence, subjective measures showed a preference for still using music

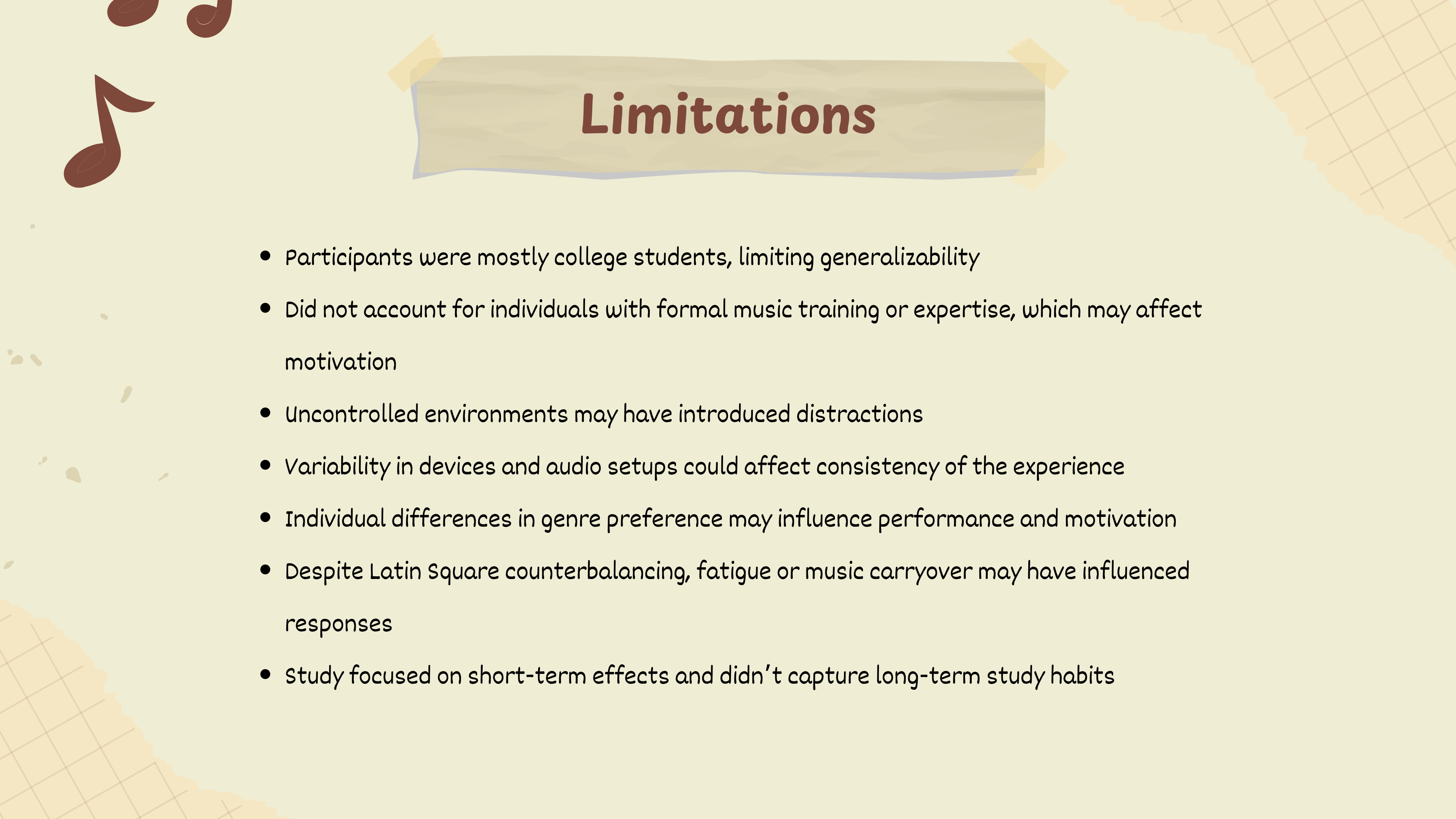


Discussion



Implications

- There was no significant differences in the conditions, likely due to insufficient data/time to accurately test our conditions
- Large variance across the data indicate no clear correlations
- Not enough time on the distractors (Indicative from high math accuracies)
- Lack of control for environment most likely influenced variance of data
- Not clear on defining motivation or creating strong enough trial to test (marginal difference in pre/post motivation test)
- People seemed to enjoy the experiment! (High correlation with motivation and mood)

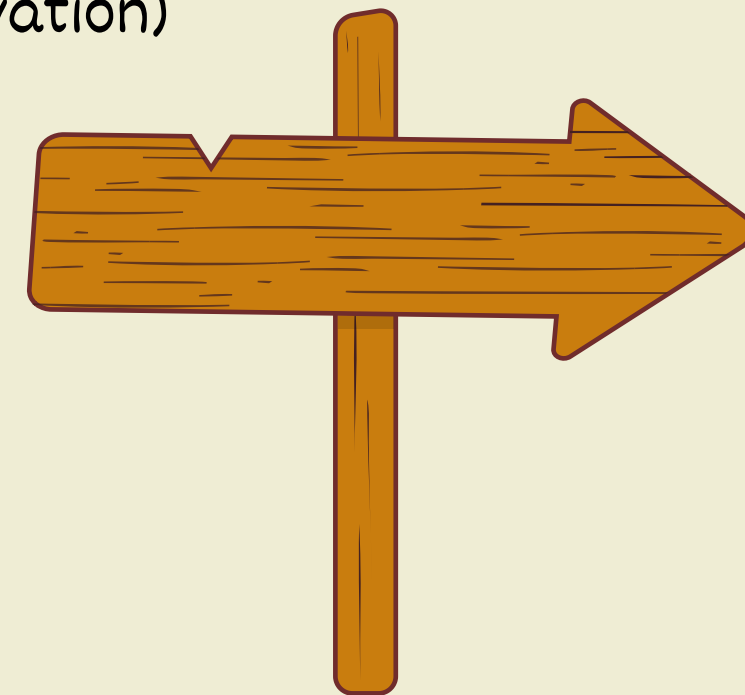


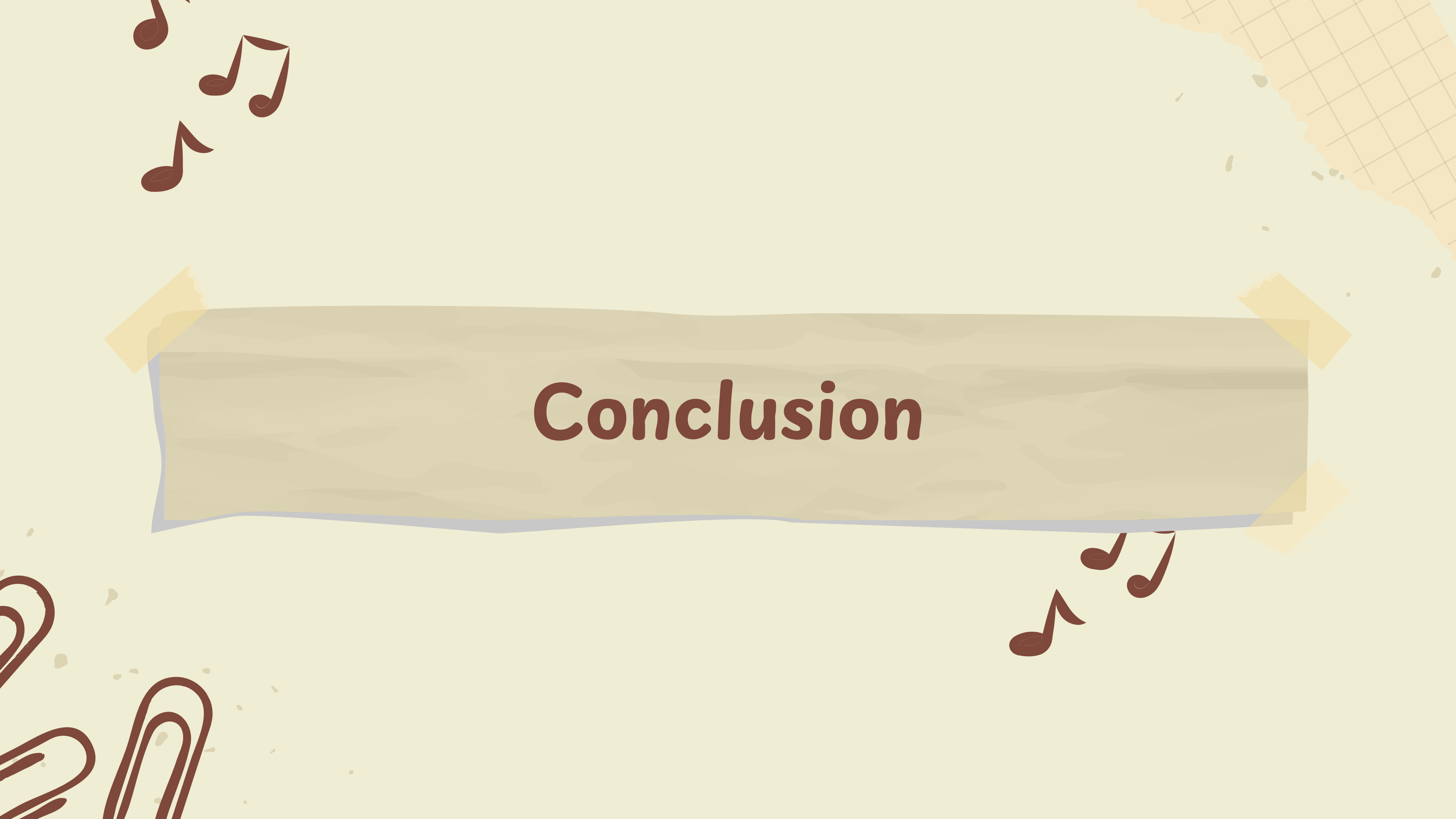
Limitations

- Participants were mostly college students, limiting generalizability
- Did not account for individuals with formal music training or expertise, which may affect motivation
- Uncontrolled environments may have introduced distractions
- Variability in devices and audio setups could affect consistency of the experience
- Individual differences in genre preference may influence performance and motivation
- Despite Latin Square counterbalancing, fatigue or music carryover may have influenced responses
- Study focused on short-term effects and didn't capture long-term study habits

Future Directions

- Larger sample size
- Doing a longer-term study that keeps up with participants over a course of days with studying and learning (longitudinal study)
- Finding out how to use the participants preferred choice of music in a structure experiment
 - Also allowing for them to do their typical study habits with music and see the effects
- Look into the addictive components of music and how this may contribute to the use of it as stimulus for motivation while studying (also mood impacts with motivation)





Conclusion

Conclusion

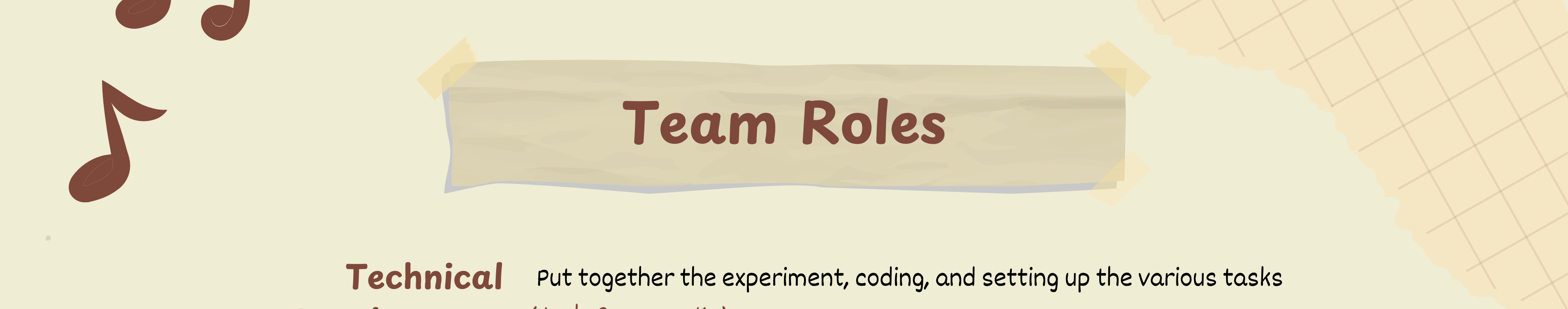
Findings

- Unable to reject null
- High variance in collected data, small usable sample size

What happened?

- Data collected in a nonstandard manner
 - Surveys sent out to friends informally instead of SONA
- Survey taken for extra credit in other classes
 - Lack of incentive to provide good data so rushed through to complete





Team Roles

Technical Development

Put together the experiment, coding, and setting up the various tasks
(Jack, Spencer, Ke)

Survey Design

Created questions for the baseline evaluation survey and the reflection question survey incorporated into the experiment
(Emily, Spencer, Jack)

Participant Coordinator

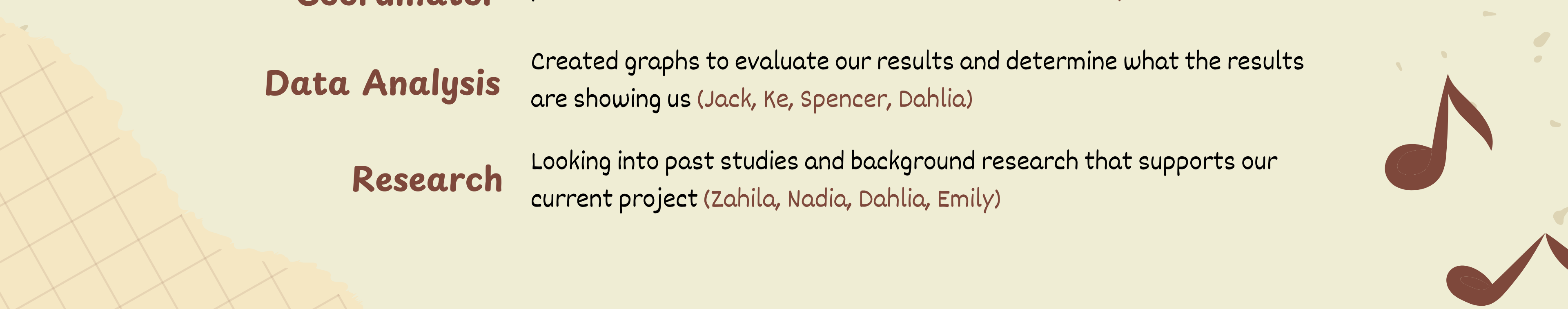
Reached out to participants and shared the experiment with professors to send to their classes (Zahila, Nadia, Emily, Jack)

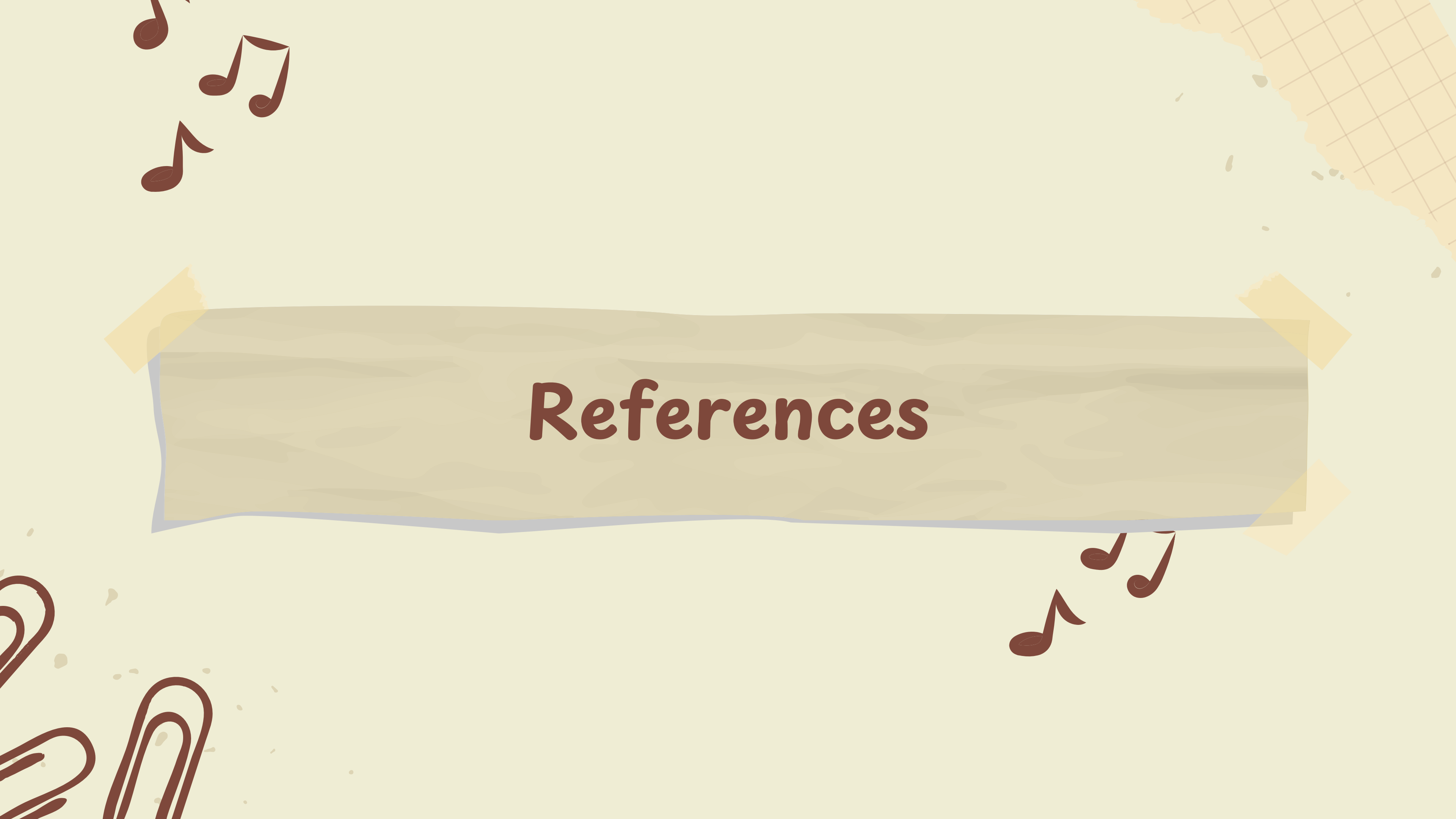
Data Analysis

Created graphs to evaluate our results and determine what the results are showing us (Jack, Ke, Spencer, Dahlia)

Research

Looking into past studies and background research that supports our current project (Zahila, Nadia, Dahlia, Emily)





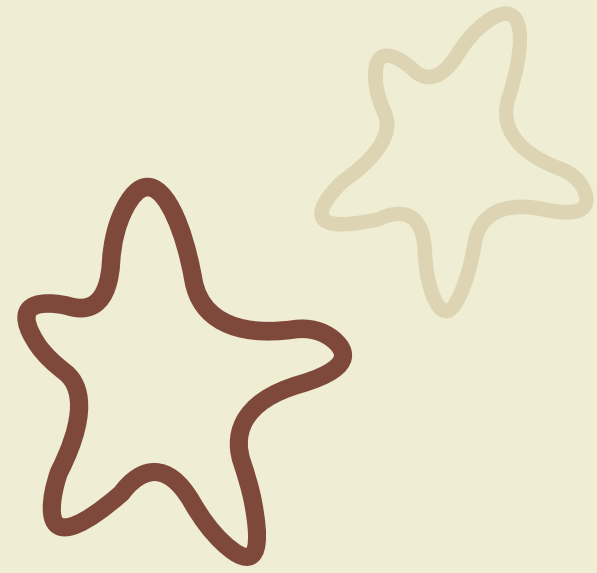
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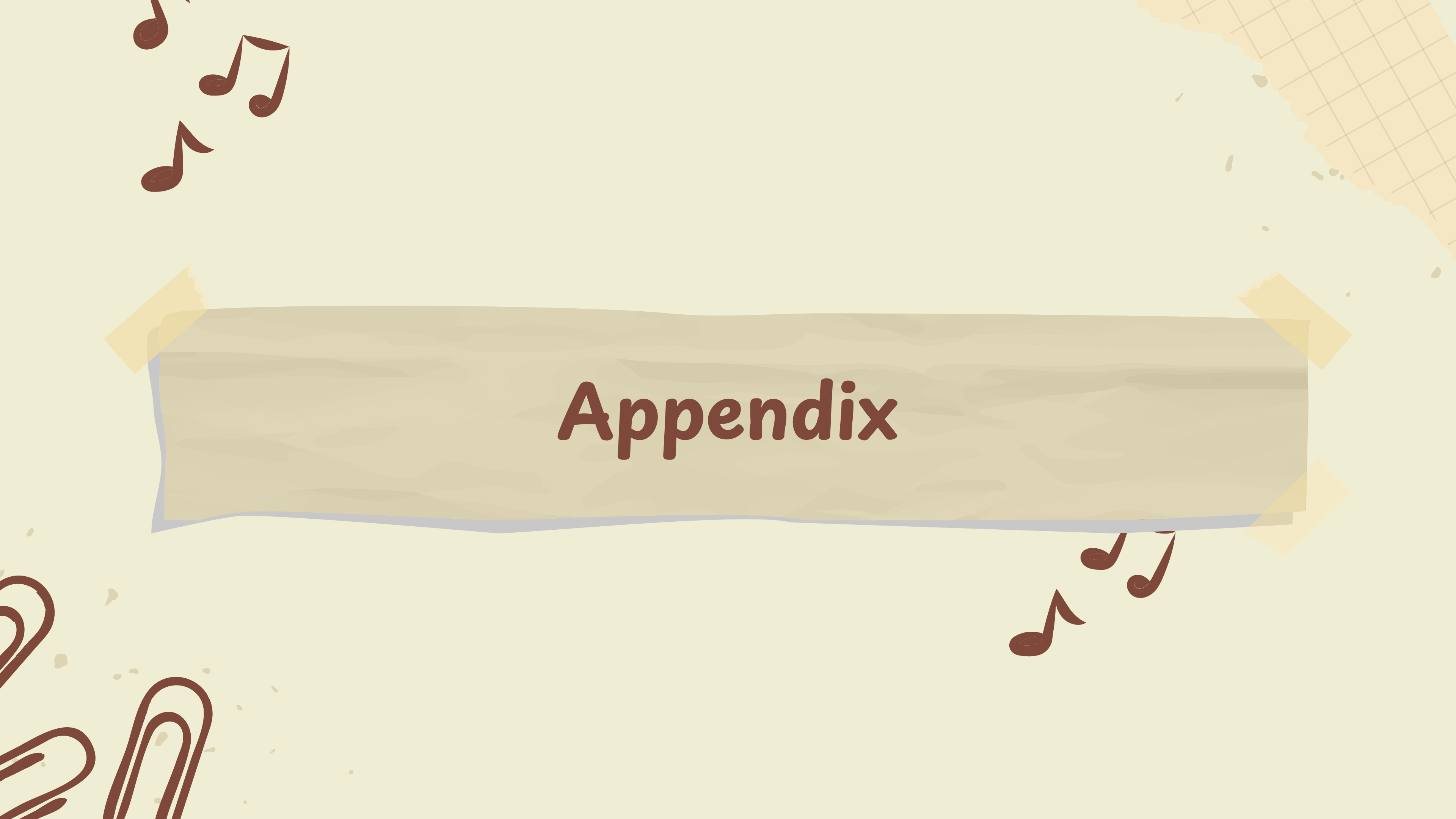
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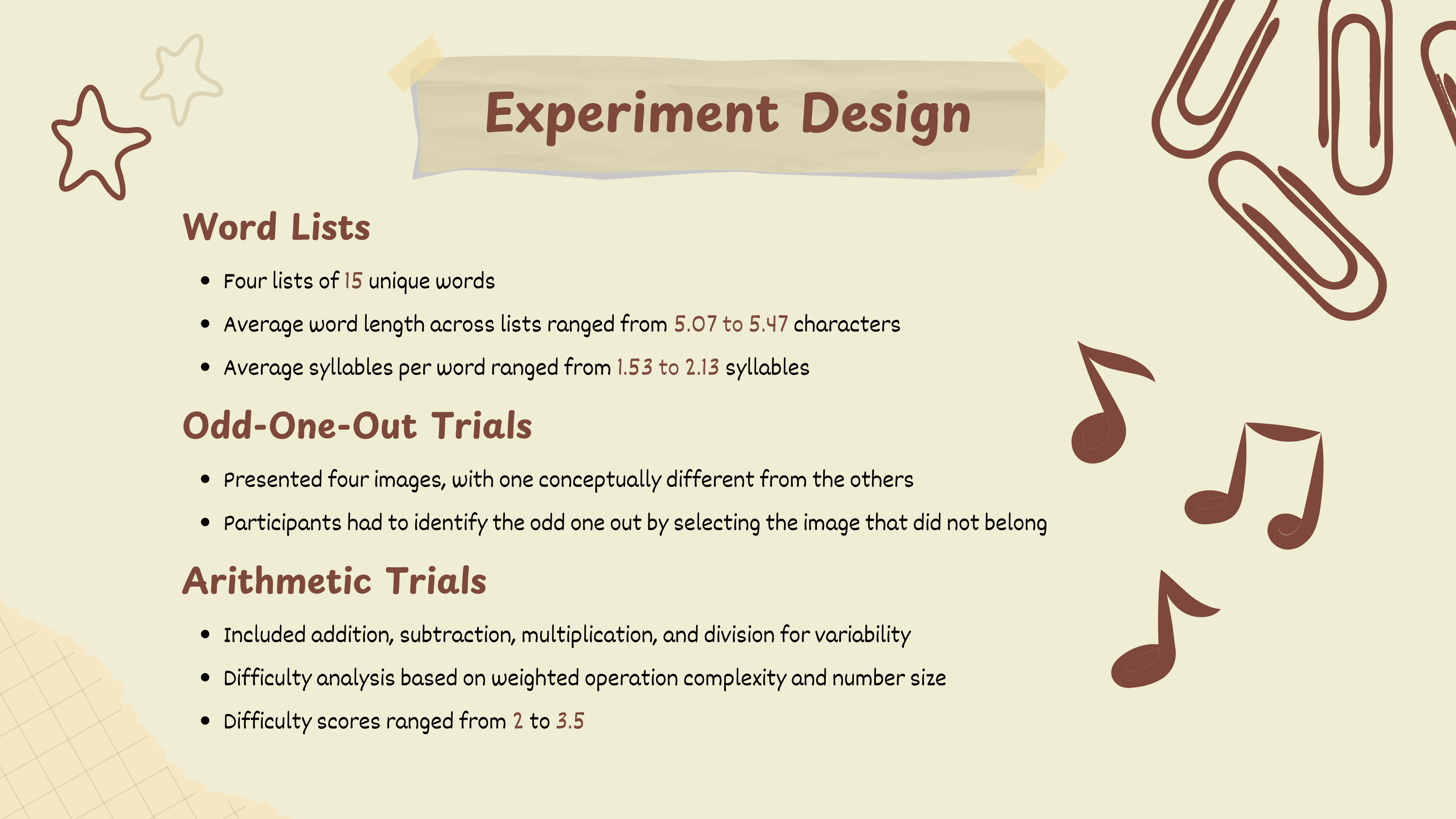
Thank You!

Any Questions??



The image features a central, horizontal, light-brown rectangular card with a slightly textured, watercolor-like appearance. The card is held in place by four yellow, torn-edge adhesive tape pieces at its corners. The word "Appendix" is written in a bold, dark brown, sans-serif font in the center of the card. The background is a light cream color. In the top-left corner, three dark brown musical notes are floating. In the bottom-left corner, two dark brown paper clips are visible. In the bottom-right corner, two more dark brown musical notes are floating. A portion of a yellow, grid-patterned object, possibly a notepad or folder, is visible in the top-right corner.

Appendix



Experiment Design

Word Lists

- Four lists of 15 unique words
- Average word length across lists ranged from 5.07 to 5.47 characters
- Average syllables per word ranged from 1.53 to 2.13 syllables

Odd-One-Out Trials

- Presented four images, with one conceptually different from the others
- Participants had to identify the odd one out by selecting the image that did not belong

Arithmetic Trials

- Included addition, subtraction, multiplication, and division for variability
- Difficulty analysis based on weighted operation complexity and number size
- Difficulty scores ranged from 2 to 3.5

Experiment Link

<https://jhcelaya.github.io/Cogs164Project/>

